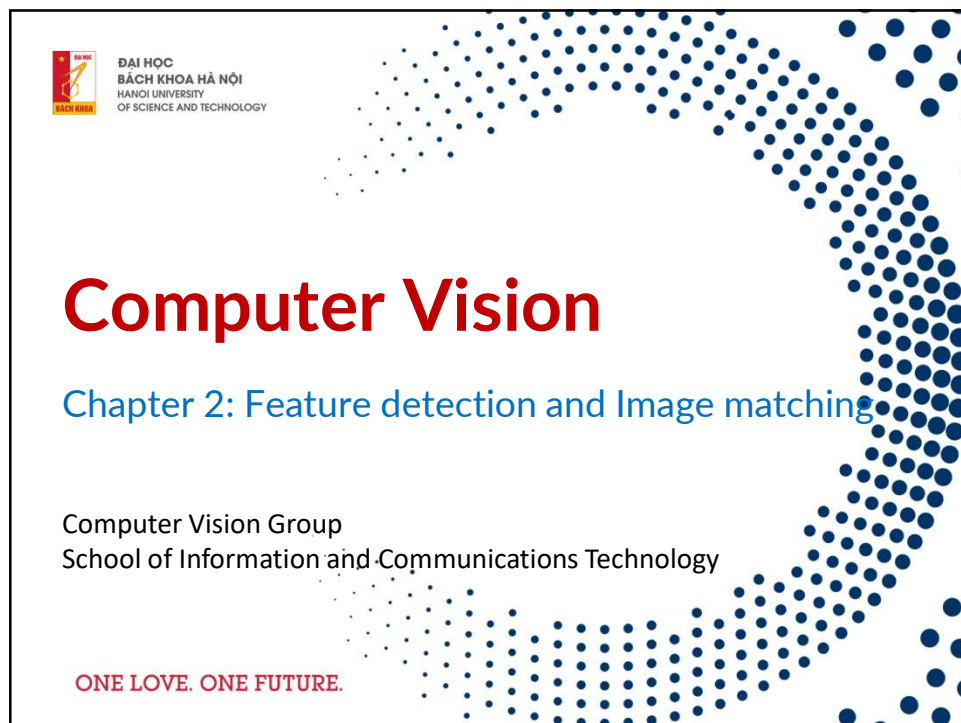




1



2

Contents

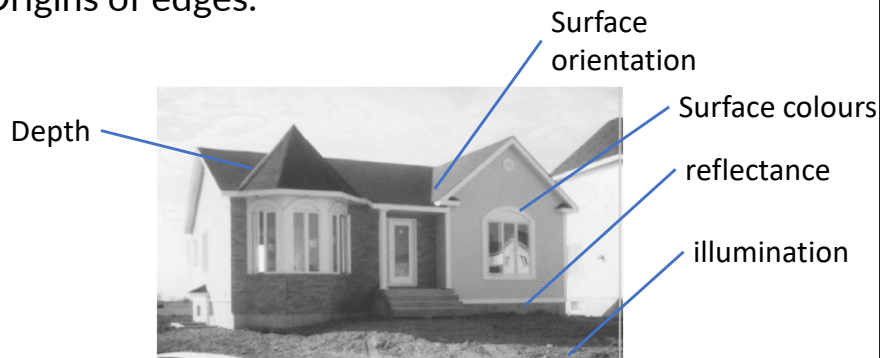
- Edge detection
 - Role of edges & general approach
 - Image gradient: sobel, prewitt,
 - Canny detector, Laplacian
 - Line detection from contour points
 - Hough transform
 - [Ransac](additional reading)
- Feature extraction
- Image Matching and Applications



3

What are edges/contours ?

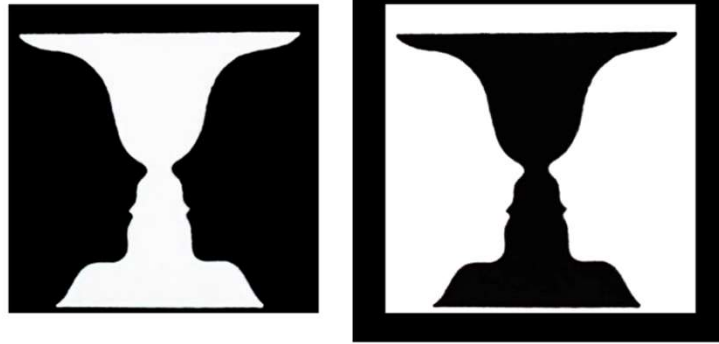
- Local changes in the images
- Typically occur on the boundary between different regions in an image
- Origins of edges:



4

Edge is important?

- What do you see ?



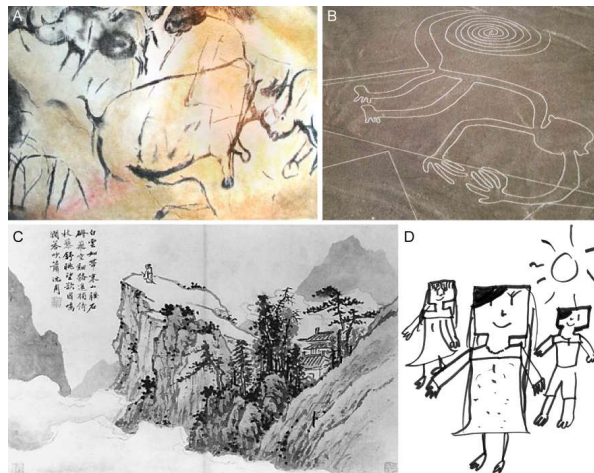
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5

5

Edge is important?

- (A) Cave painting at Chauvet, France, about 30,000 B.C.;
 (B) Aerial photograph of the picture of a monkey as part of the Nazca Lines geoglyphs, Peru, about 700 – 200 B.C.;
 (C) Shen Zhou (1427-1509 A.D.): Poet on a mountain top, ink on paper, China;
 (D) Line drawing by 7-year old I. Lleras (2010 A.D.).



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152 Biederman

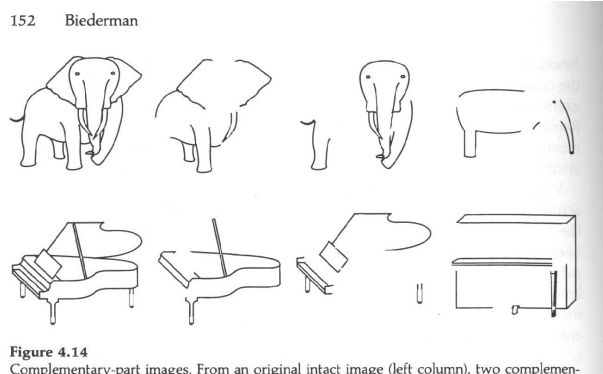


Figure 4.14

Complementary-part images. From an original intact image (left column), two complemen-

Can we recognize these objects?



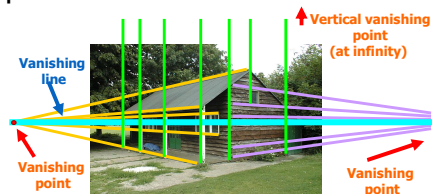
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8

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Edge detection

- Goal: Identify sudden changes (discontinuities) in an image
 - Intuitively, most semantic and shape information from the image can be encoded in the edges
 - More compact than pixels
- Why?
 - Extract information, recognize objects
 - Recover geometry and viewpoint



Source: J. Hayes

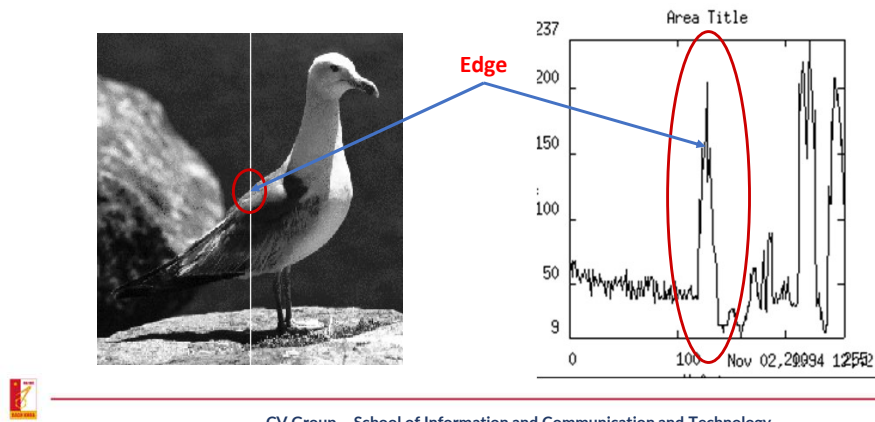
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9

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How to find edges?

- **Intensity profile** of an image is the set of intensity values taken from regularly spaced points along a line segment or multi-line path in an image

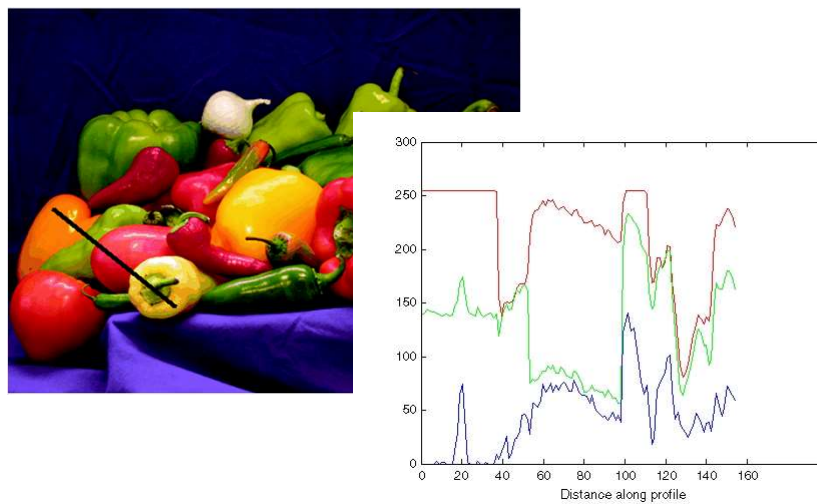


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Image profile



<https://www.mathworks.com/help/images/intensity-profile-of-images.html>

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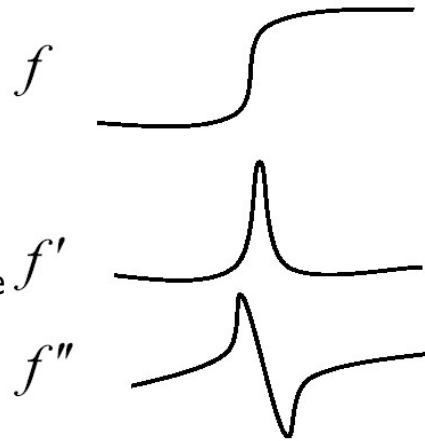
11

How to find edges?

- An edge is a place of rapid change in the image intensity function

- Extrema of 1st derivative

- Zero-crossing of 2nd derivative



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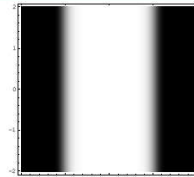
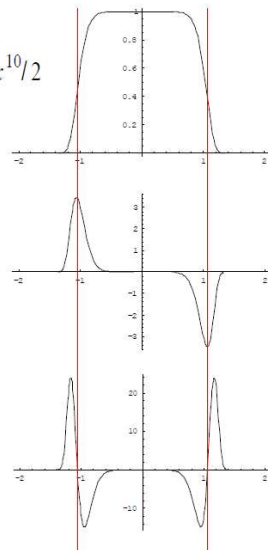
12

How to find edges?

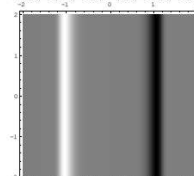
$$f(x, y) = e^{-x^{10}/2}$$

$$\frac{\partial f}{\partial x}$$

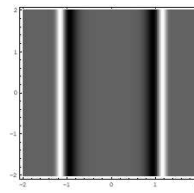
$$\frac{\partial^2 f}{\partial x^2}$$



Image



The first derivative



The second derivative



Source : Caroline Rougier. Traitement d'images (IFT2730). Univ. de Montréal.

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Image gradients



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Derivatives

- Derivative in 1D:

$$\frac{df}{dx} = \lim_{\Delta x \rightarrow 0} \frac{f(x) - f(x - \Delta x)}{\Delta x} = f'(x) = f_x$$

- Discrete derivative in 1D

$$\frac{df}{dx} = \frac{f(x) - f(x-1)}{1} = f'(x)$$

$$\frac{df}{dx} = f(x) - f(x-1) = f'(x)$$



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1D discrete derivate filters

- Backward filter:

$$\frac{df}{dx} = f(x) - f(x-1) = f'(x) \quad [0 \quad 1 \quad -1]$$

- Forward:

$$\frac{df}{dx} = f(x) - f(x+1) = f'(x) \quad [-1 \quad 1 \quad 0]$$

- Central:

$$\frac{df}{dx} = f(x+1) - f(x-1) = f'(x) \quad [1 \quad 0 \quad -1]$$



1D discrete derivate example

- Backward filter: $[0 \quad 1 \quad -1]$

$$f(x) = 10 \quad 15 \quad 10 \quad 10 \quad 25 \quad 20 \quad 20 \quad 20$$

$$f'(x) = 0 \quad 5 \quad -5 \quad 0 \quad 15 \quad -5 \quad 0 \quad 0$$

$$\begin{array}{lcl} f(x) : & 0 & 0 \quad 0 \quad 0 \quad 0 \quad 50 \quad 50 \quad 50 \quad 50 \quad 50 \\ f'(x) : & 0 & 0 \quad 0 \quad 0 \quad 0 \quad 50 \quad 0 \quad 0 \quad 0 \quad 0 \end{array}$$



Discrete derivative in 2D

Given function $f(x, y)$

Gradient vector $\nabla f(x, y) = \begin{bmatrix} \frac{\partial f(x, y)}{\partial x} \\ \frac{\partial f(x, y)}{\partial y} \end{bmatrix} = \begin{bmatrix} f_x \\ f_y \end{bmatrix}$

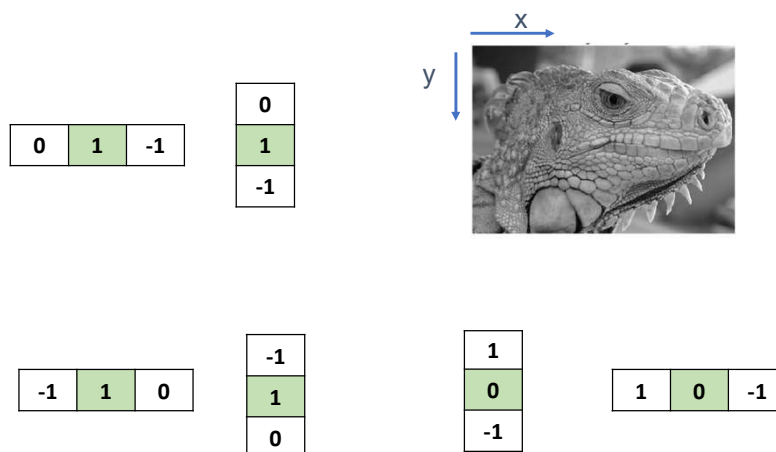
Gradient magnitude $|\nabla f(x, y)| = \sqrt{f_x^2 + f_y^2}$

Gradient direction $\theta = \tan^{-1} \frac{f_x}{f_y}$



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Discrete derivative filters wrt. x and y



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Image I (9 x 8)

0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	100	100	100	100	100	100
0	0	0	100	100	100	100	100	100
0	0	0	100	100	100	100	100	100
0	0	0	100	100	100	100	100	100
0	0	0	100	100	100	100	100	100
0	0	0	100	100	100	100	100	100

M_x : Derivative filter wrt x

1	0	-1
---	---	----

M_y : Derivative filter wrt y

1
0
-1

$I_y = I * M_y$ = the 1st derivative wrt y

0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	100	100	100	100	100
0	0	100	100	100	100	100
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0

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3x3 image gradient filters

$$\frac{1}{3} \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix} \quad \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix}$$

Derivative in x direction

Derivative in y direction



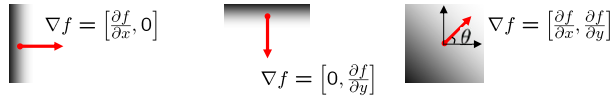
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Image gradient

- The gradient of an image:

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right]$$



- The **gradient vector** points in the direction of most rapid increase in intensity
- The **gradient direction** is given by $\theta = \tan^{-1} \left(\frac{\partial f}{\partial y} / \frac{\partial f}{\partial x} \right)$
 - how does this relate to the direction of the edge?
- The **edge strength** is given by the **gradient magnitude**

$$\|\nabla f\| = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2} \approx |\partial f / \partial x| + |\partial f / \partial y|$$

Source: Steve Seitz



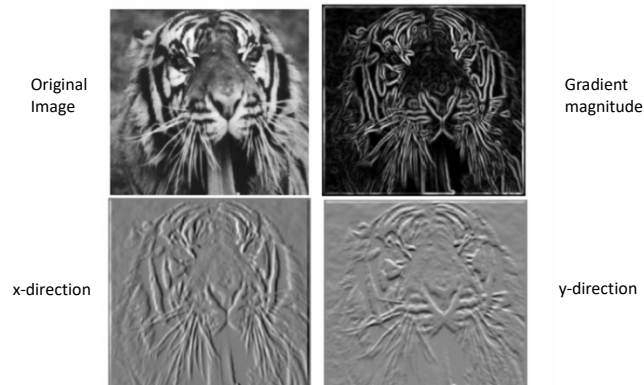
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Example

- Which one is the gradient in the x-direction? How about y-direction?



CS231: Lecture 3 Juan Carlos Niebles and Ranjay Krishna

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Edge detector

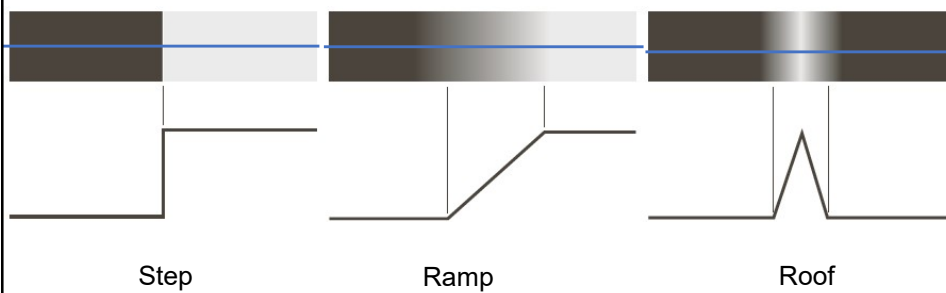


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Type of edges



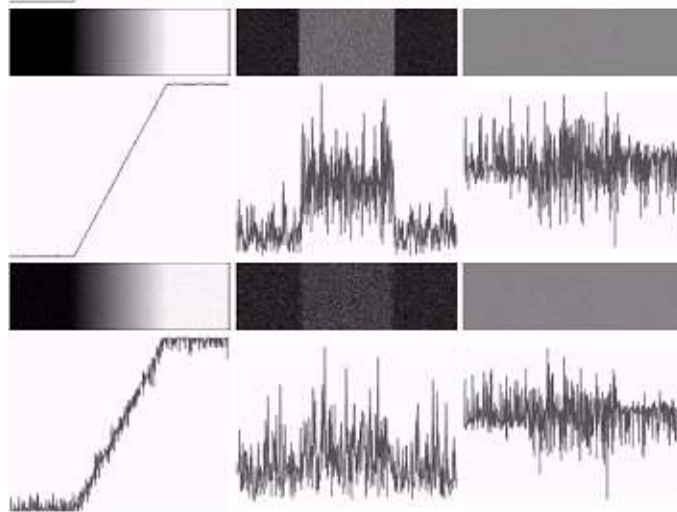
Source : Gonzalez and Woods. *Digital Image Processing 3ed.* Prentice-Hall, 2008.

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Effects of noise



Source : Gonzalez and Woods. *Digital Image Processing 3ed.* Prentice-Hall, 2008.

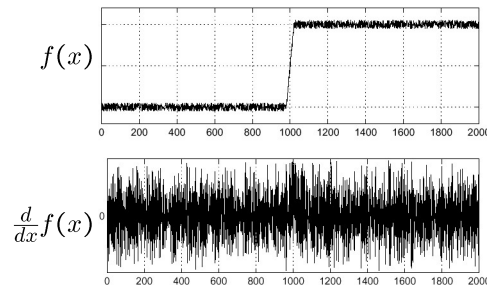
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Effects of noise

- Consider a single row or column of the image
 - Plotting intensity as a function of position gives a signal



Where is the edge?

Source: S. Seitz

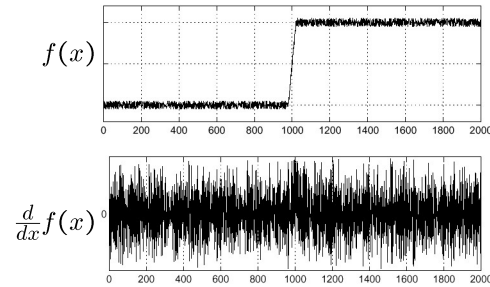
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Effects of noise

- Solution: smoothing the image



Where is the edge?



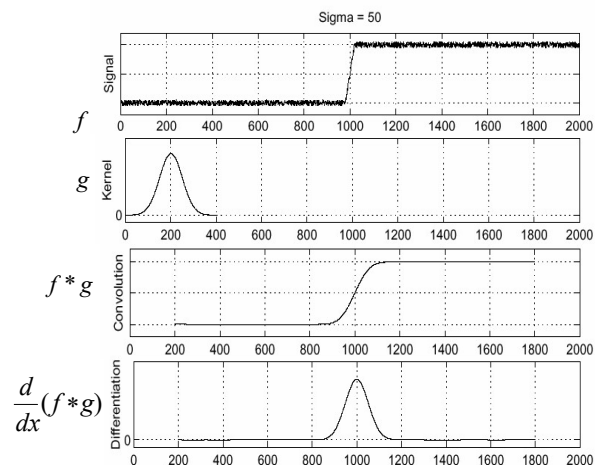
Source: S. Seitz

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Solution: smooth first



- To find edges, look for peaks in $\frac{d}{dx}(f * g)$



Source: S. Seitz

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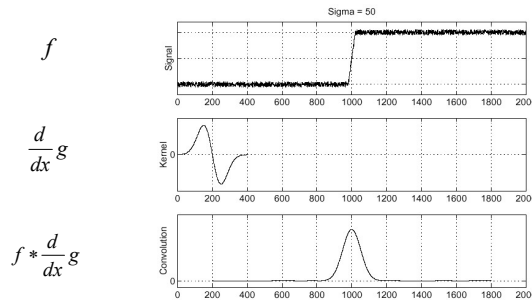
31

Derivative theorem of convolution

- This theorem gives us a very useful property:

$$\frac{d}{dx}(f * g) = \frac{df}{dx} * g = f * \frac{dg}{dx}$$

- This saves us one operation:



Source: S. Seitz



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Sobel Operator

- Gaussian Smoothing + differentiation

$$G_x = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \end{bmatrix}$$

Gaussian
smoothing

differentiation

$$G_y = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$$

➔ Less sensible to noise



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Prewitt Operator

- Mean smoothing + differentiation

$$Gx = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} [1 \ 0 \ -1]$$

$$Gy = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} [1 \ 1 \ 1]$$

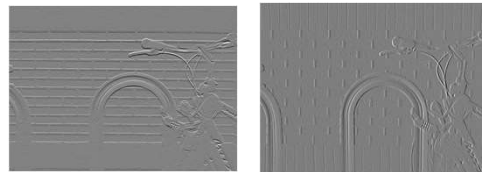
→ Less sensible to noise



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Simple edge detector with 1st derivative

- Convolve the original image with 2 kernels to calculate approximations of the derivatives



- Compute the gradient magnitude

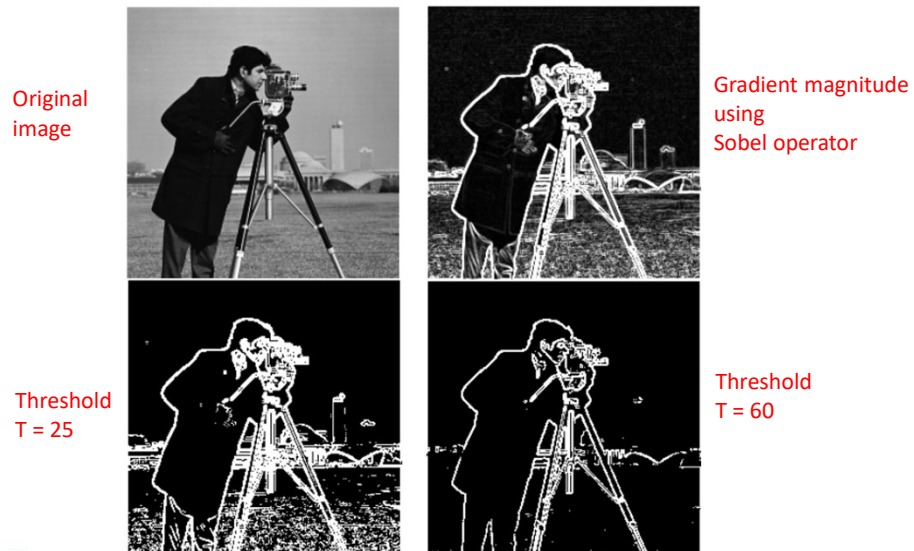


- Thesholding:
 - choose edges to be the pixel above a threshold T



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Simple edge detector with 1st derivative



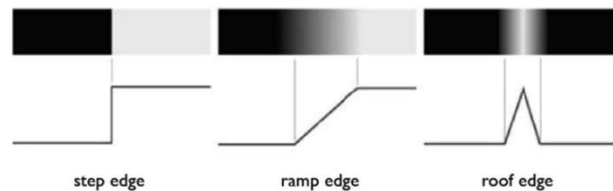
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Problems

- Poor localization (multiple adjacent pixels)
- Thresholding value favors certain directions over others
 - Can miss oblique edges more than horizontal or vertical edges → False negatives

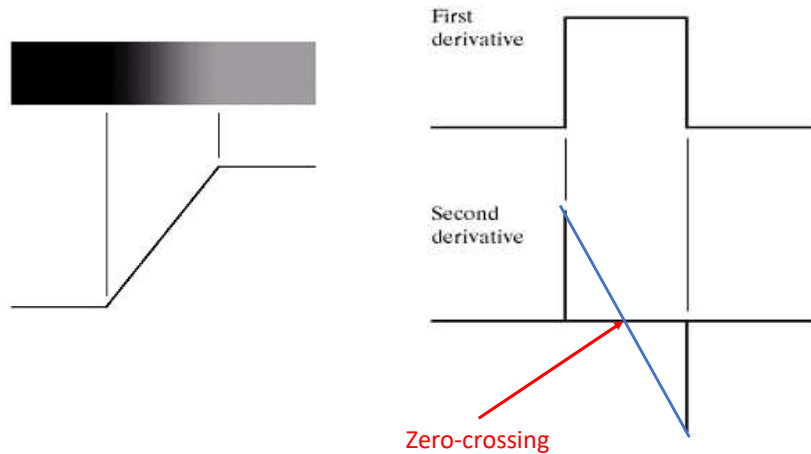


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Edge detector with 2nd derivative



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Edge detector with 2nd derivative

- 2nd derivative with Laplacian filter:
 - convolution the image with one of 2 filters

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 1 & 1 & 1 \\ 1 & -8 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

- Edge detection
 - Compute the 2nd derivative of the images
 - Find the zero-crossing pixels → edges



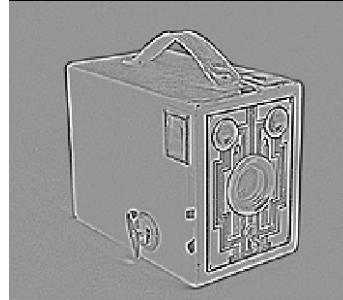
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Edge detector with 2nd derivative

Image



Laplacian



- Single response
- Sensible to noise



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Laplace of Gaussian (LoG)

- Laplace is sensitive to noise → smoothing image before convolution with the laplacian filter
 $I * G * L = I * (G * L) \rightarrow G * L: \text{LoG filter}$
- Second derivative = Image * LoG

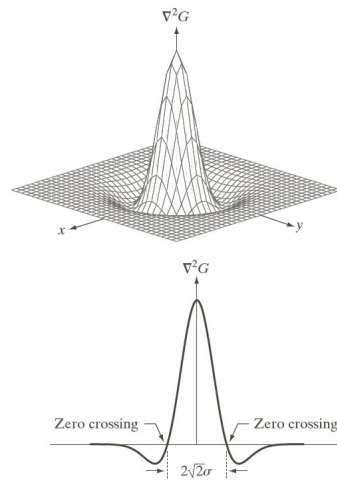
$$\text{LoG}(x, y) = \frac{\partial^2 G}{\partial x^2} + \frac{\partial^2 G}{\partial y^2}$$

$$\text{LoG}(x, y) = -\frac{1}{\pi\sigma^4} \left[1 - \frac{x^2 + y^2}{2\sigma^2} \right] e^{-\frac{x^2 + y^2}{2\sigma^2}}$$



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Laplace of Gaussian (LoG)



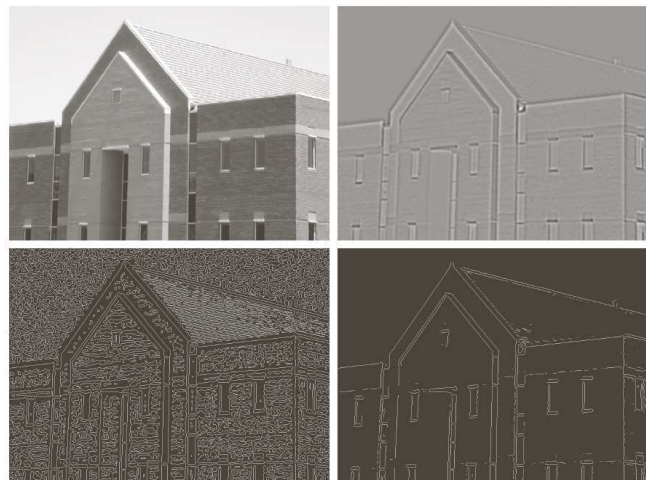
a b
c d

FIGURE 10.21

(a) Three-dimensional plot of the *negative* of the LoG. (b) Negative of the LoG displayed as an image. (c) Cross section of (a) showing zero crossings. (d) 5×5 mask approximation to the shape in (a). The negative of this mask would be used in practice.



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a b
c d

FIGURE 10.22

(a) Original image of size 834×1114 pixels, with intensity values scaled to the range $[0, 1]$. (b) Results of Steps 1 and 2 of the Marr-Hildreth algorithm using $\sigma = 4$ and $n = 25$. (c) Zero crossings of (b) using a threshold of 0 (note the closed-loop edges). (d) Zero crossings found using a threshold equal to 4% of the maximum value of the image in (b). Note the thin edges.

Called also Marr–Hildreth (name of two authors)

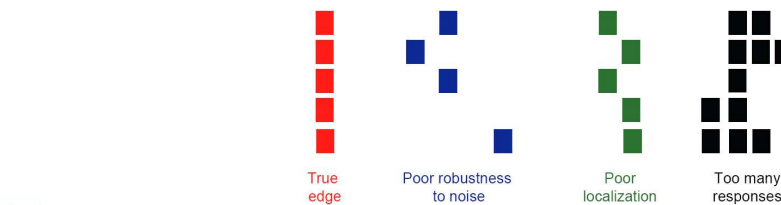


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“Optimal” edge detector

- **Criteria:**

- **Good detection:** the optimal detector must minimize the probability of false positives (detecting spurious edges caused by noise), as well as that of false negatives (missing real edges)
- **Good localization:** the edges detected must be as close as possible to the true edges
- **Single response:** the detector must return one point only for each true edge point; that is, minimize the number of local maxima around the true edge



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Canny detector

- This is probably the **most widely used** edge detector in computer vision
- Theoretical model: step-edges corrupted by additive Gaussian noise
- Canny has shown that the first derivative of the Gaussian closely approximates the operator that **optimizes the product of signal-to-noise ratio and localization**



J. Canny, [A Computational Approach To Edge Detection](#), IEEE Trans. PAMI, 8:679-714, 1986.

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Canny detector

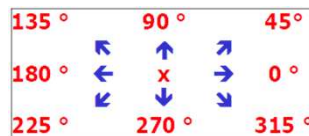
- Optimal:
 - Detection: **weak edges detected**
 - Good location: **close to the real edges**
 - Unique response: **edge thickness = 1**



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Canny detector: Steps

- 1) Apply a gaussian filter on the image
 - Lowpass filter to **remove noise**
- 2) Compute the **gradient intensity** in the image
 - Sobel filter in X and Y
 - Compute the magnitude $|G| = |G_x| + |G_y|$
- 3) Compute image **gradient direction**
 - Gradient direction $\theta = \arctan (G_y / G_x)$
 - Round directions using multiples of 45°



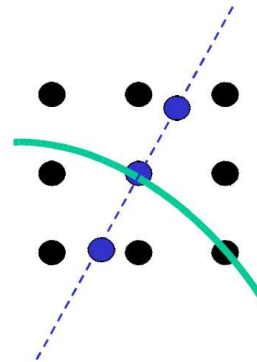
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Canny detector: Steps

4) Non-maxima suppression

- If the gradient magnitude of a pixel (x,y) is inferior to the one of its 2 neighbors along the gradient direction

→ set this magnitude for (x,y) to zero

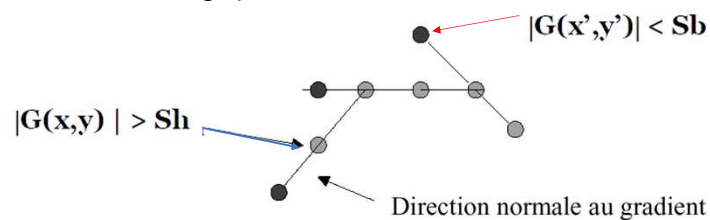


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Canny detector: Steps

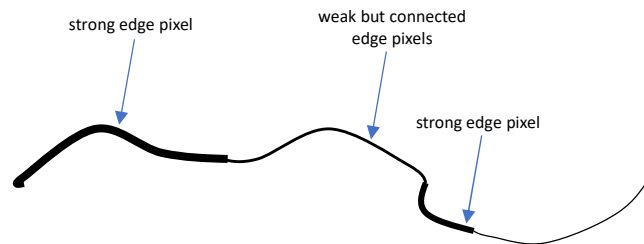
5) Edge thresholding (hysteresis)

- Using two thresholds: a threshold high (S_h) and a threshold low (S_b)
- For each pixel in the gradient magnitude:
 - IF $\text{magnitude}(x,y) < S_b$, THEN set the pixel to zero (non-edge)
 - IF $\text{magnitude}(x,y) > S_h$, THEN the edge is an edge
 - IF $S_b \leq \text{magnitude}(x,y) \leq S_h$, THEN the pixel is an edge IF it is connected to another edge pixel



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Hysteresis thresholding



Source: S. Seitz

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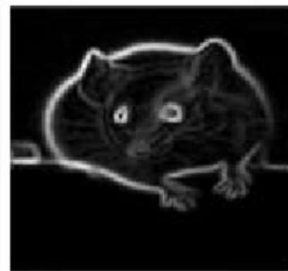
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Canny detector

Input image



Sobel



Non-maxima suppression



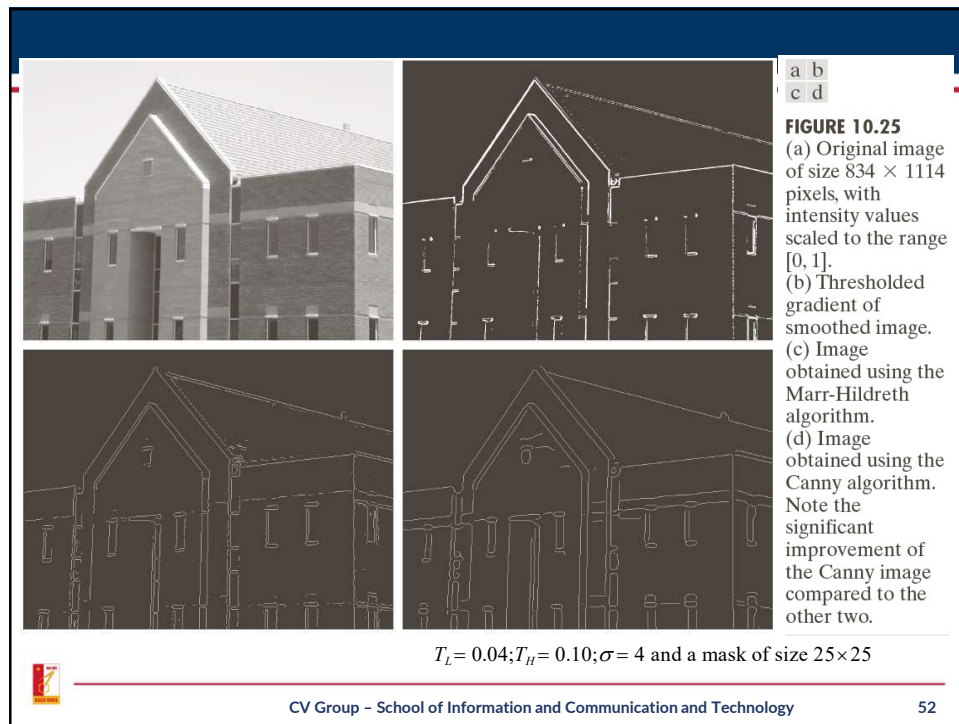
Thresholding



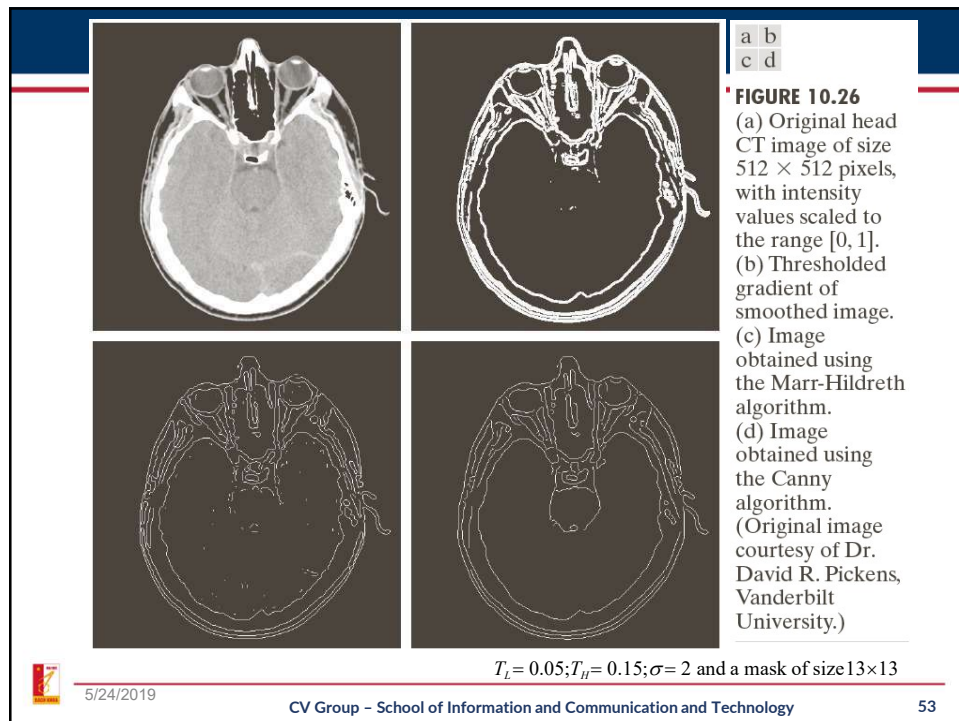
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Plan

- Edge detection
 - Role of edges & general approach
 - Image gradient: sobel, prewitt,
 - Canny detector, Laplacian
 - Edge linking: line detection from contour points
 - Hough transform
 - [Ransac](additional reading)
- Feature extraction
- Image Matching and Applications



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Hough transform

- The Hough transform (HT)
 - can be used to detect lines
 - was introduced in 1962 (Hough 1962) and first used to find lines in images a decade later (Duda 1972)
 - Goal: to find the location of lines in images.
- **Caveat:** Hough transform can *detect lines, circles* and other *structures ONLY* if their *parametric equation is known*
- It can give **robust detection** under *noise* and *partial occlusion*



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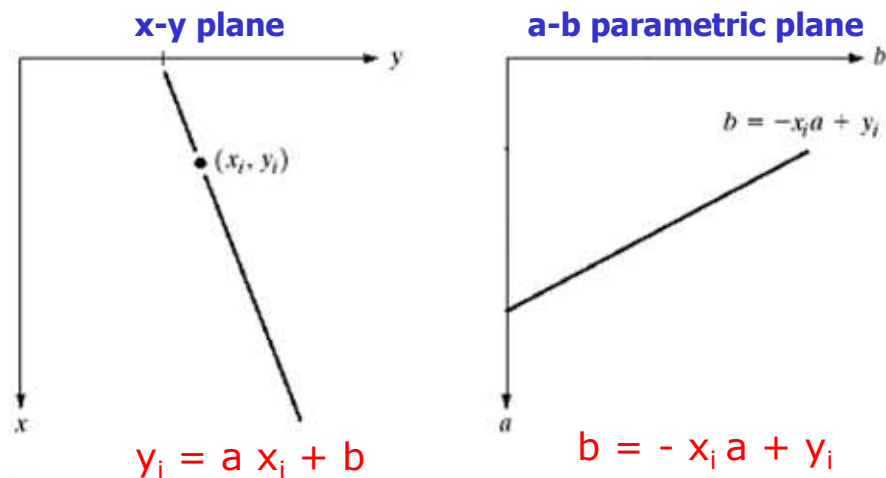
Hough transform

- **Global** approach to detect continuous edges
 - *From the x-y plane to the parametric plane a-b*
- **x-y plane**
 - $y_i = a x_i + b$
 - an infinity of lines going through one (x_i, y_i) pair
 - one sole line for the (a,b) pair
- **a-b parametric plane**
 - $b = -x_i a + y_i$
 - one sole line for the (x_i, y_i) pair
 - an infinity of lines going through one (a,b) pair



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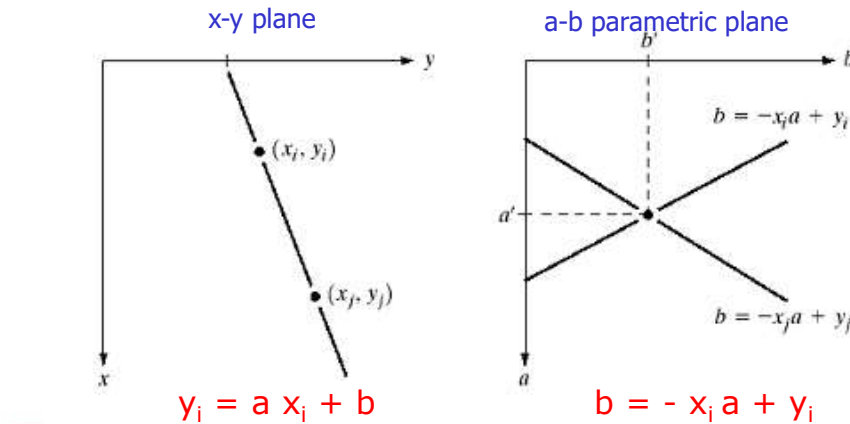
x-y plane vs a-b plane



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Line vs Points

All the points (x,y) on a line in the x - y plane are going through one sole point (a', b') in the a - b parametric plane



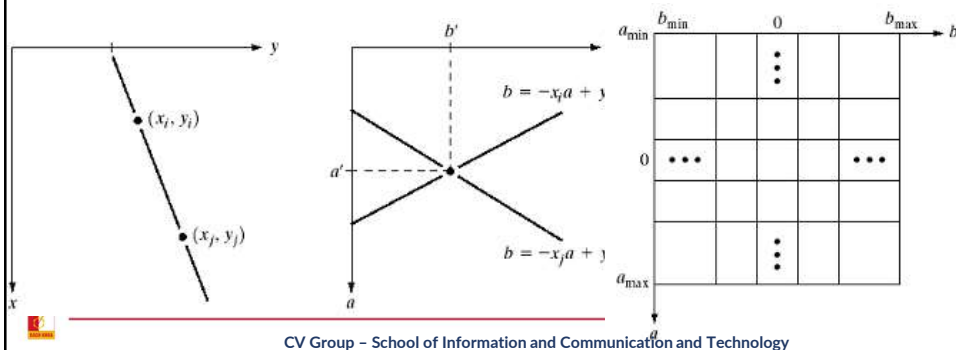
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Main idea for the Hough transform

- Accumulation cells - Matrix (a,b)
- Build an **voting image**
 - each point is voting for a particular line
- The lines receiving more votes are kept



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Computing the Hough transform

- We compute the **contour points** of the input image
 - *Sobel, Prewitt, Canny, ...*
- For each contour point, we compute a line **(a,b)**
 - *Result is one line in the a-b plane for each pixel (x,y)*
- The maximum peaks in the a-b parametric plane show the lines with the maximum of points in the x-y plane
 - *The points indicating line crossing in the a-b plane show the real lines existing in the x-y plane*

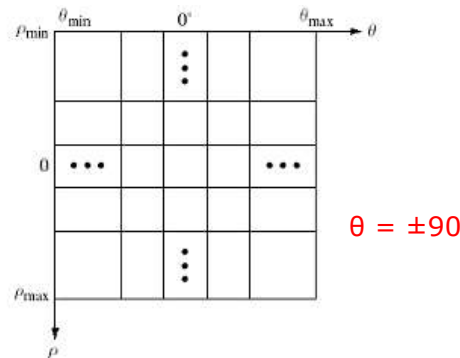
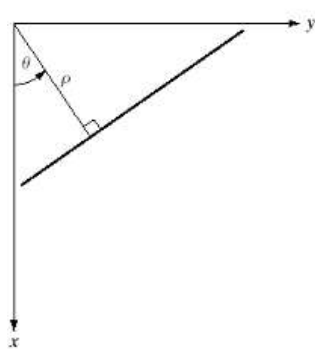


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Problem with the (a,b) space

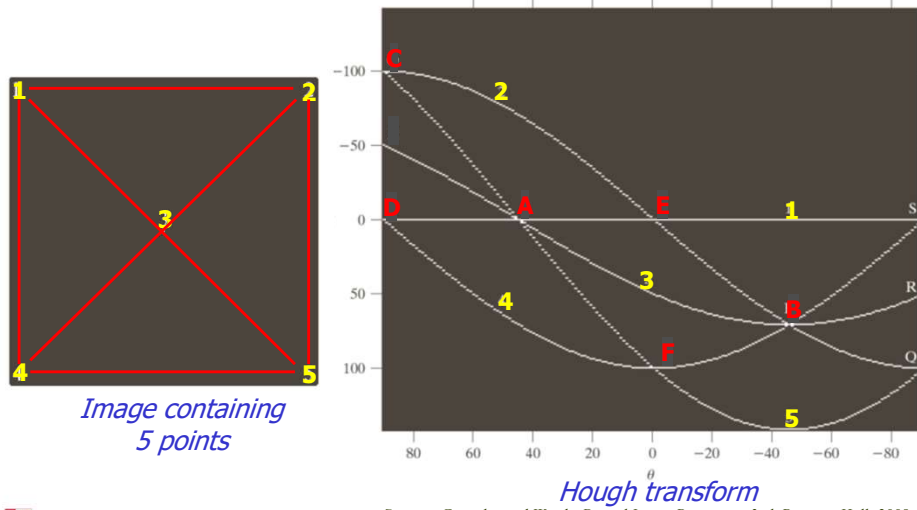
- **Problem:** for a vertical line, $a = \infty$!
- **Solution:** representing using polar coordinates (ρ, θ)

$$\rho = x \cos \theta + y \sin \theta$$



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Example with 5 points

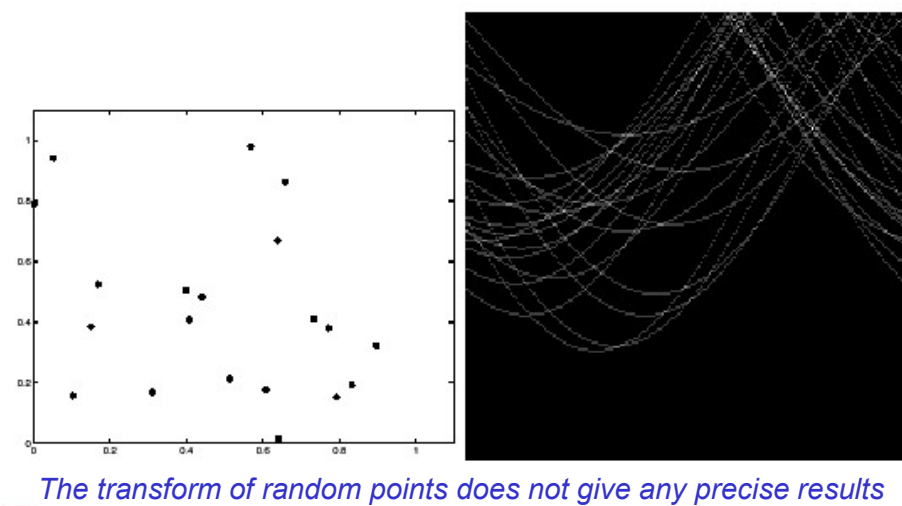


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Hough transform (random points)



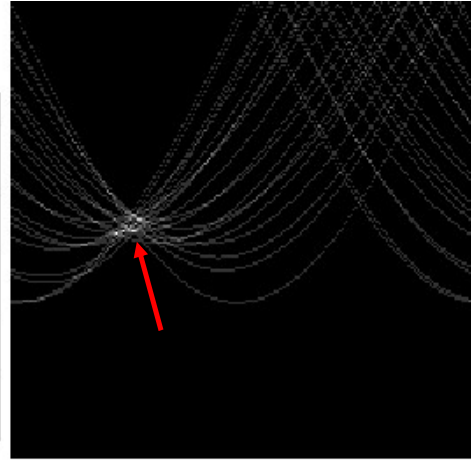
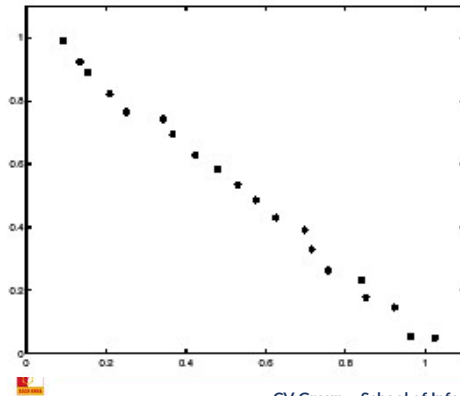
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Hough transform (straight line)

The transform for aligned points result in a line detection



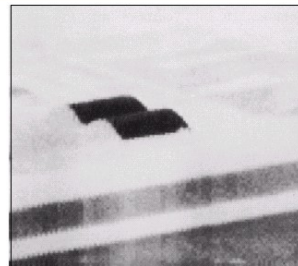
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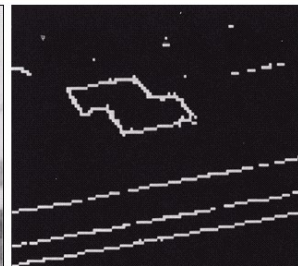
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Example

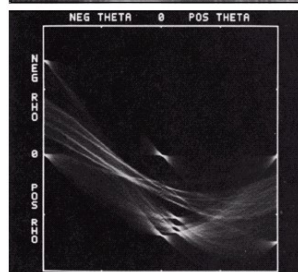
Image



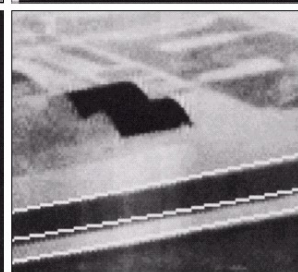
Gradient



Hough



Final



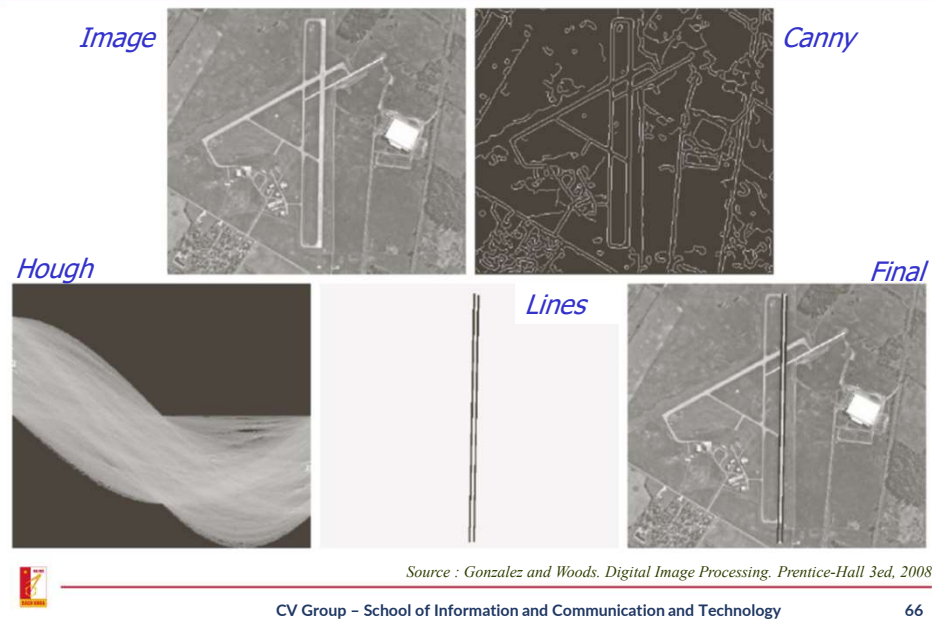
Source : Gonzalez and Woods, Digital Image Processing, Prentice-Hall, 2002.

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Other example



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Ransac (additional reading)

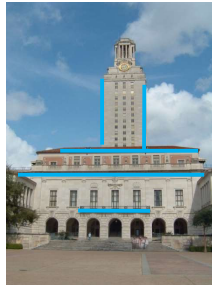
- A model fitting method:
 - A learning technique to estimate parameters of a model by random sampling of observed data
 - Used for :
 - Line detection
 - Correspondance problem (matching between 2 sets of features)
 - ...



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Example: Line Fitting

- Why fit lines?
 - Many objects characterized by presence of straight lines



Slide credit: Kristen Grauman



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Fitting as search in parametric space

- Choose a **parametric model** to represent a set of features
- Membership criterion is **not local**
 - Can't tell whether a point belongs to a given model just by looking at that point.
- Three main questions:
 - **What model** represents this set of features best?
 - **Which of several model instances** gets which feature?
 - How **many model instances** are there?
- Computational **complexity** is important
 - It is infeasible to examine every possible set of parameters and every possible combination of features



Source: L. Lazebnik

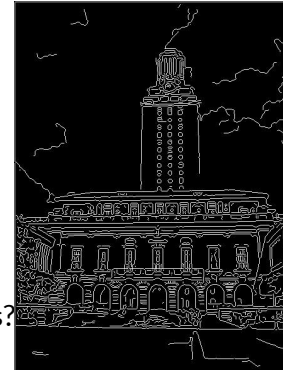
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Difficulty of Line Fitting

- Extra edge points (clutter), multiple models:
 - Which points go with which line, if any?
- Only some parts of each line detected, and some parts are missing:
 - How to find a line that bridges missing evidence?
- Noise in measured edge points, orientations:
 - How to detect true underlying parameters?



Slide credit: Kristen Grauman

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Voting

- It's **not feasible to check all combinations** of features by fitting a model to each possible subset.
- Voting is a general technique where we let **the features vote for all models** that are compatible with it.
 - Cycle through features, cast votes for model parameters.
 - Look for model parameters that receive a lot of votes.
- **Noise & clutter** features will cast votes too, *but* typically their votes should be inconsistent with **the majority of "good" features**.
- Ok if some features not observed, as model can span multiple fragments.



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RANSAC (Fischler & Bolles 1981)

- **RAN**dom **SA**mples **C**onsensus
- Approach: we want to avoid the impact of outliers, so let's look for "inliers", and use only those
- Intuition: if an outlier is chosen to compute the current fit, then the resulting line won't have much support from rest of the points.



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RANSAC (Fischler & Bolles 1981)

RANSAC loop:

1. Randomly **select a seed group** of points on which to base transformation estimate (e.g., a group of matches)
 2. **Compute transformation** from seed group
 3. Find **inliers** to this transformation
 4. If the number of inliers is **sufficiently large, re-compute least-squares estimate** of transformation on all of the inliers
- Keep the transformation with the largest number of inliers



Slide credit: Kristen Grauman

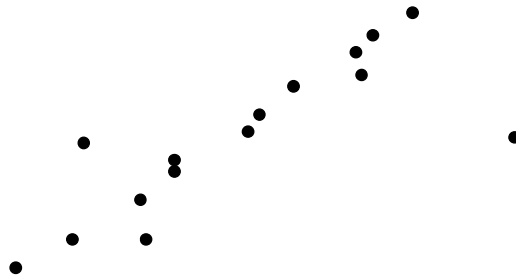
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RANSAC Line Fitting Example

- Task: Estimate the best line
 - *How many points do we need to estimate the line?*



Slide credit: Kristen Grauman

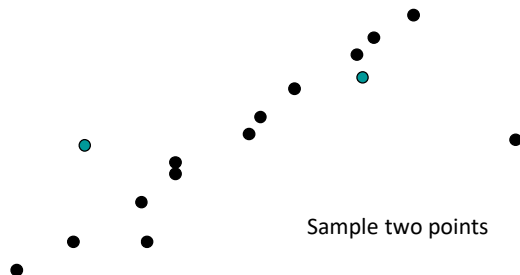
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RANSAC Line Fitting Example

- Task: Estimate the best line



Slide credit: Kristen Grauman

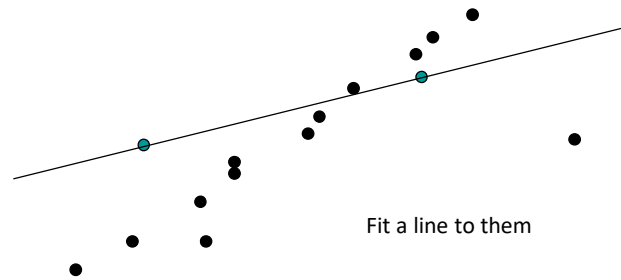
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RANSAC Line Fitting Example

- Task: Estimate the best line



Slide credit: Kristen Grauman

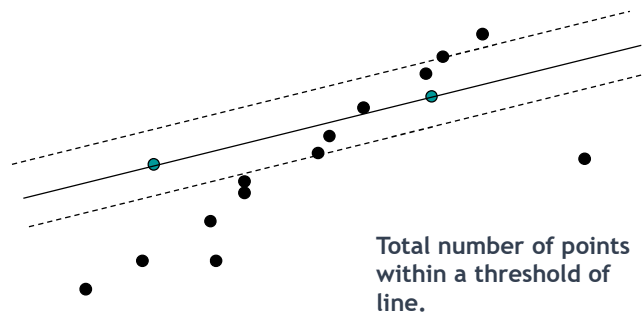
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RANSAC Line Fitting Example

- Task: Estimate the best line



Slide credit: Kristen Grauman

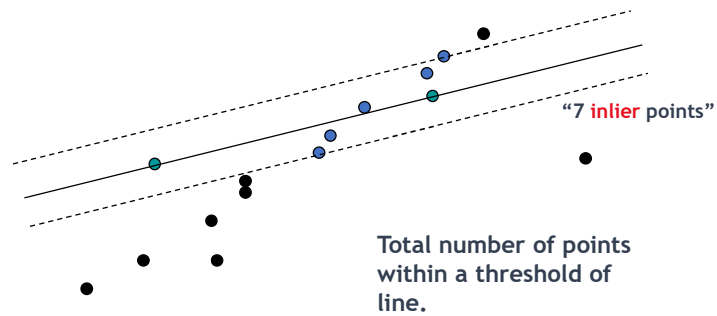
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RANSAC Line Fitting Example

- Task: Estimate the best line



Slide credit: Kristen Grauman

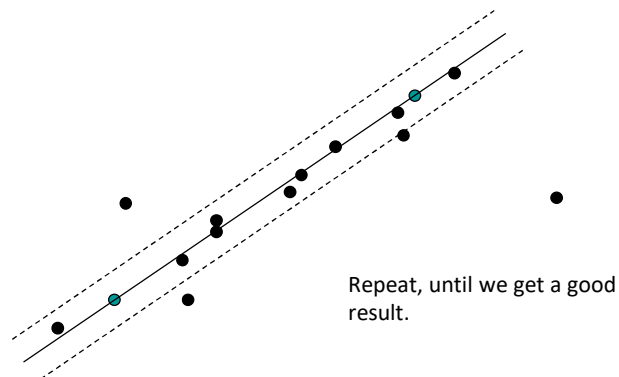
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RANSAC Line Fitting Example

- Task: Estimate the best line



Slide credit: Kristen Grauman

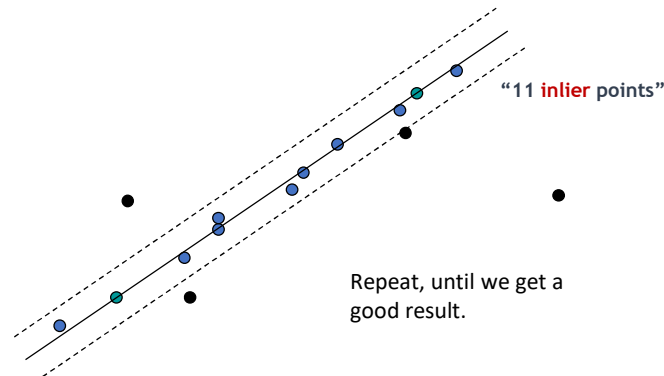
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RANSAC Line Fitting Example

- Task: Estimate the best line



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Algorithm 15.4: RANSAC: fitting lines using random sample consensus

Determine:

- n — the smallest number of points required
- k — the number of iterations required
- t — the threshold used to identify a point that fits well
- d — the number of nearby points required to assert a model fits well

Until k iterations have occurred

Draw a sample of n points from the data uniformly and at random

Fit to that set of n points

For each data point outside the sample

Test the distance from the point to the line against t ; if the distance from the point to the line is less than t , the point is close

end

If there are d or more points close to the line then there is a good fit. Refit the line using all these points.

end

Use the best fit from this collection, using the fitting error as a criterion



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RANSAC: How many samples?

- How many samples are needed?
 - Suppose w is fraction of inliers (points from line)
 - n points needed to define hypothesis (2 for lines)
 - k samples chosen
 - p : desired probability that we get a good sample

- Prob. that a single sample of n points is correct:

$$w^n$$

- Prob. that all k samples fail is:

$$(1 - w^n)^k$$

⇒ Choose k high enough to keep this below desired failure rate $(1-p)$.



Slide credit: David Lowe

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RANSAC: Computed k ($p=0.99$)

Sample size n	Proportion of outliers						
	5%	10%	20%	25%	30%	40%	50%
2	2	3	5	6	7	11	17
3	3	4	7	9	11	19	35
4	3	5	9	13	17	34	72
5	4	6	12	17	26	57	146
6	4	7	16	24	37	97	293
7	4	8	20	33	54	163	588
8	5	9	26	44	78	272	1177

p : desired probability that we get a good sample



Slide credit: David Lowe

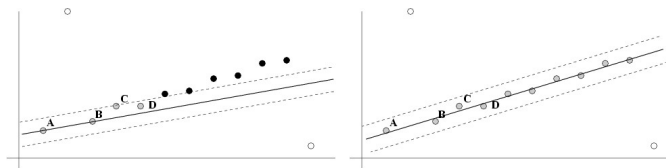
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After RANSAC

- RANSAC divides data into **inliers** and **outliers** and yields estimate computed from minimal set of inliers.
- Improve **this initial estimate with estimation over all inliers** (e.g. with standard least-squares minimization).
- But this may change inliers, so alternate fitting with re-classification as inlier/outlier.



Slide credit: David Lowe

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RANSAC: Pros and Cons

- **Pros:**
 - General **method suited for a wide range of model fitting** problems
 - **Easy** to implement and easy to calculate its failure rate
- **Cons:**
 - **Only handles a moderate percentage of outliers** without cost blowing up
 - Many real problems have **high rate of outliers** (but sometimes selective choice of random subsets can help)
- A voting strategy, the Hough transform can handle high percentage of outliers



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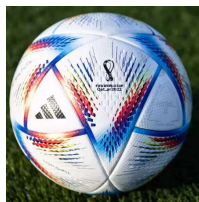
Plan

- Edge detection
 - General approach
 - Image gradient, Canny detector, Laplacian
 - Line detection from contour points (Hough transform)
- Feature Extraction
 - Global features
 - Local features
- Image matching and Applications



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Feature extraction: what and why?



Matrix of integers:

- Redundant information
- Many of discrete numbers

?

A ball?
An official match ball
for Qatar 2022?

Decision model

A (compact) representation that
describes well the object/image that
is used to distinguish object from one
another.

(called **Features**)



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Feature extraction: what and why?



Can we GO
or STOP?



Useful feature: **color**



Player vs
Ball?



Useful feature:
shape

<https://www.theguardian.com/>



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Good feature?

- A good feature will help us recognize an object in all the ways it may appear
 - Identifiable
 - Easily tracked and compared
 - Consistent across different scales, lighting conditions, and viewing angles
 - Still visible in noisy images or when only part of an object is visible



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Local features vs Global features

- Two types of features are extracted from the image:
 - local and global features (descriptors)
- **Global feature:**
 - Describe the **image as a whole** to generalize the entire object
 - Include contour representations, shape descriptors, and texture features
 - Examples: Invariant Moments (Hu, Zernike), Histogram Oriented Gradients (HOG), PHOG, and Co-HOG,...
- **Local feature:**
 - the local features **describe the image patches** (key points in the image) of an object
 - represents the texture/color in an image patch
 - Examples: SIFT, SURF, LBP, BRISK, MSER and FREAK, ...



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Feature extraction

- **Global features**
 - **Color / Shape / Texture**
- Local features



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Global features?

How to distinguish these objects?



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Types of features

- Contour representation, Shape features
- Color descriptors
- Texture features



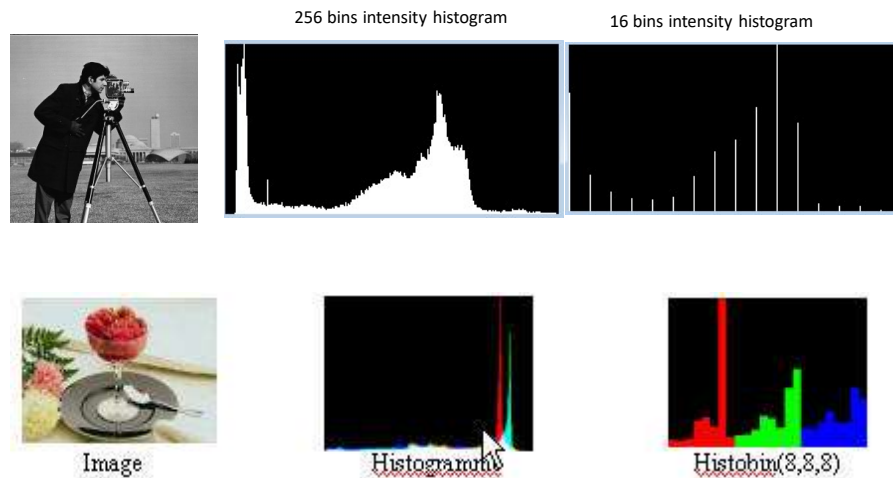
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Color features

- Histogram



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Distance / Similarity

- L1 ou L2 (euclidian) distances are often used

$$d_{L1}(H, G) = \sum_{i=1}^N |h_i - g_i|$$

- Histogram intersection

$$\cap(H, G) = \frac{\sum_i \min(h_i, g_i)}{\sum_i g_i}$$

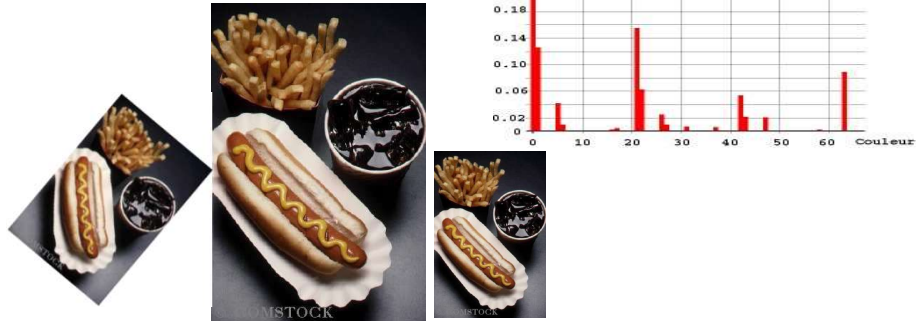


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Advantages of histogram

- Invariant to basic geometric transformations:

- Rotation
- Translation
- Zoom (Scaling)



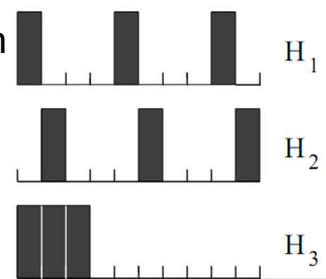
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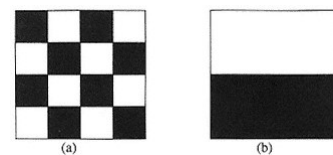
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Some inconveniences

- The **similarity between colors in adjacent colors** (bin) is not taken into account



- The **spatial distribution** of pixel values is **not considered**: 2 different images may have the same histogram



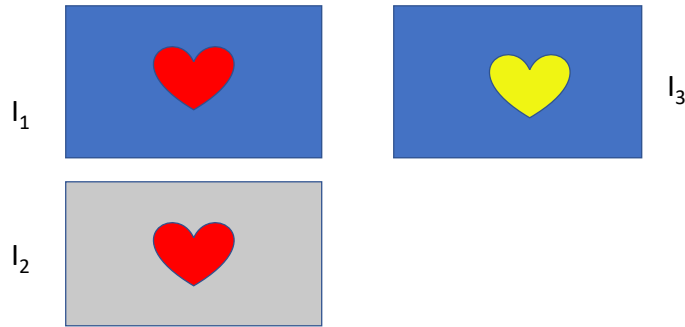
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Some inconveniences

- **Background effect:** $d(I_1, I_2) \neq d(I_1, I_3)$



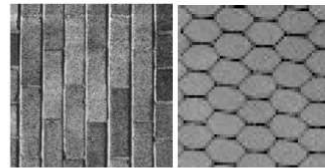
- **Color representation dependency** (color space), device dependency, ...



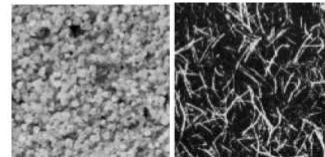
Texture features

- A **texture** can be defined as
 - a region with variation of intensity
 - as a spatial organization of pixels

A texture can be **periodic** (a pattern that repeats itself) ...



...or **non-periodic** (no pattern, more disorganized)



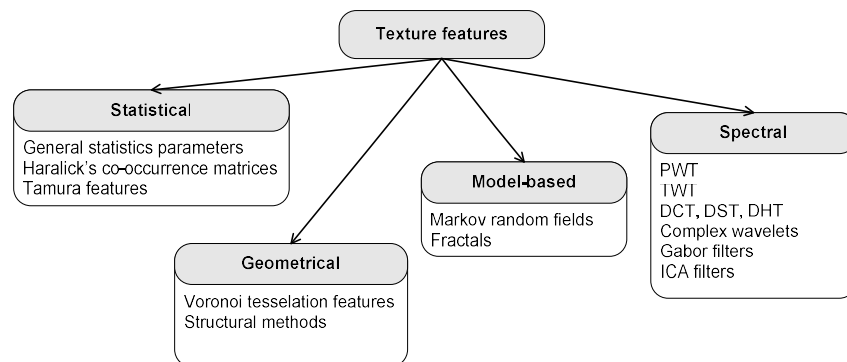
Texture features

- There are several methods for analyzing textures:
 - First order statistics
 - Statistics on histogram
 - Co-occurrence matrices
 - Searching patterns
 - Frequential analysis
 - Gabor filter
 - ...
- The most difficult is to find a good representation (good parameters) for each texture



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Texture features



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First order statistics

- Histogram-based: mean, variance, skewness, kurtosis, energy, entropy, ...
- Given an image with n pixels:
 - $h(i)$ represents the number of occurrences of the i -th gray level.
 - $L-1$: maximum gray level
 - $P(i)$: normalized histogram

$$P(i) = \frac{h(i)}{n}$$



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First order statistics

$$\mu = \sum_{i=0}^{L-1} (i \cdot P(i))$$

Mean = image brightness

$$\sigma^2 = \sum_{i=0}^{L-1} (i - \mu)^2 \cdot P(i)$$

Variance measures the deviation of gray levels from the Mean

$$s = \frac{1}{\sigma^3} \sum_{i=0}^{L-1} (i - \mu)^3 \cdot P(i)$$

Skewness is a measure of the degree of histogram asymmetry around the Mean

$$k = \frac{1}{\sigma^4} \sum_{i=0}^{L-1} (i - \mu)^4 \cdot P(i) - 3$$

Kurtosis is a measure of the histogram sharpness

$$H = - \sum_{i=0}^{L-1} P(i) \log P(i)$$

Entropy is a measure of randomness

$$E = \sum_{i=0}^{L-1} (P(i))^2$$

Energy measures the smoothness

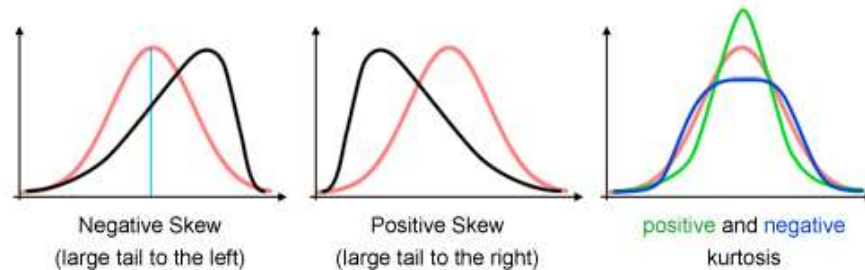


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First order statistics

- Skewness vs. kurtosis (red: normal distribution with mean = 5, variance = 4)

© www.scratchapixel.com



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GLCM (Gray Level Co-occurrence Matrices)

- Features generated from the first-order statistics:
 - Provide information related to the gray-level distribution of the image
 - But do not give any information about the relative positions of the various gray levels within the image

→ Gray Level Co-occurrences matrix that measures second-order image statistics



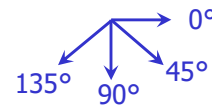
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GLCM (Gray Level Co-occurrence Matrices)

- GLCM describes how frequently two pixels with gray-levels c_i, c_j appear in the image, separated by a distance d in direction β
 - Haralick features
- Matrix of size $L \times L$
 - L is the number of gray level in the image (256x256)
 - We often reduce that number to 8x8, 16x16 or 32x32
- Many matrices, one for each distance and direction:
 - Relative distance (in pixel numbers) d : 1, 2, 3 (4, ...)
 - Relative direction β : $0^\circ, 45^\circ, 90^\circ, 135^\circ$ (, ...)
- Processing time can be very long



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GLCM (Gray Level Co-occurrence Matrices)

- the number of pairs of pixels (p_1, p_2) where
- p_2 is a neighbor of p_1
 - p_1 with gray level c_i and p_2 with gray level c_j

$$CM_{d,\beta}(c_i, c_j) = \frac{\text{card}(\{p_1, p_2 \mid I(p_1) = c_i, I(p_2) = c_j, N_{d,\beta}(p_1, p_2) = \text{true}\})}{\text{card}(\{p_1, p_2 \mid N_{d,\beta}(p_1, p_2) = \text{true}\})}$$

- the number of all pairs (p_1, p_2) where p_2 is the neighbor of p_1 (used to normalize the GLCM)

Card(.) : the cardinality of a set

$$N_{d,\beta}(p_1, p_2) = \text{true}$$

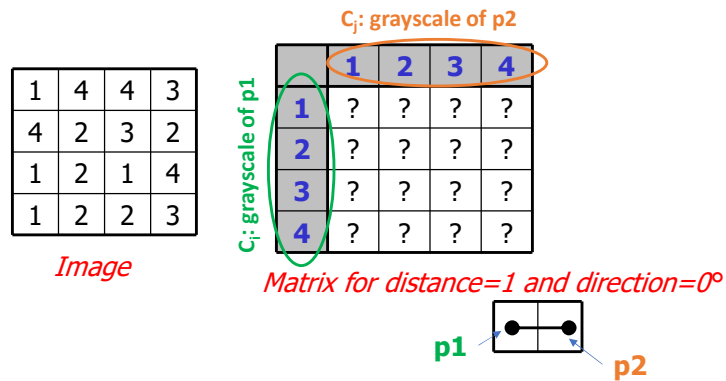
Pixel p_2 is a neighbor of pixel p_1 at a distance d in direction β



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GLCM (Gray Level Co-occurrence Matrices)

- Example on how to compute these matrices:



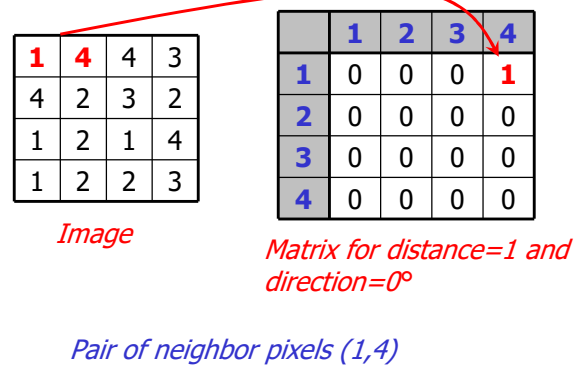
We loop over the image and for each pair of pixels following the given distance and orientation, we increment the co-occurrence matrix



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GLCM (Gray Level Co-occurrence Matrices)

- Example on how to compute these matrices:



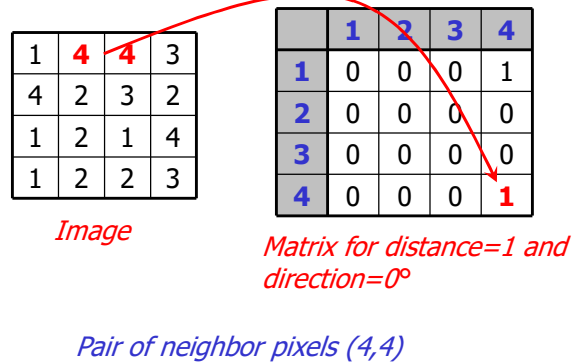
Pair of neighbor pixels (1,4)



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GLCM (Gray Level Co-occurrence Matrices)

- Example on how to compute these matrices:



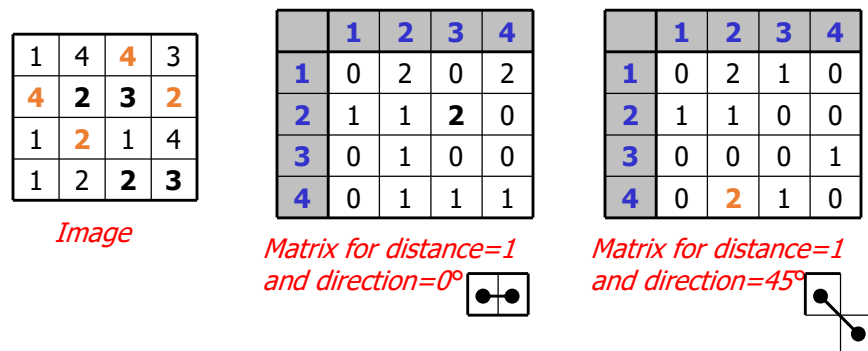
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GLCM (Gray Level Co-occurrence Matrices)

- Example on how to compute these matrices (final):



...and so on for each matrix (several matrices at the end)



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GLCM (Gray Level Co-occurrence Matrices)

- Most important/popular parameters computed from GLCM:

$$ASM = \sum_i \sum_j CM^2(i, j)$$

Angular Second Moment (ASM):
minimal when all elements are equal

$$Entropy = - \sum_i \sum_j CM(i, j) \log(CM(i, j))$$

a measure of chaos,
maximal when all elements are equal

$$Contrast = \sum_i \sum_j (i - j)^2 CM(i, j)$$

small values when big elements
are near the main diagonal

$$idm = \sum_i \sum_j \frac{1}{1 + (i - j)^2} CM(i, j)$$

idm (inverse differential moment) has small values
when big elements are far from the main diagonal



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GLCM (Gray Level Co-occurrence Matrices)

- Haralick features:
 - For each GLCM, we can compute up to 14 parameters characterizing the texture, of which the most important : mean, variance, energy, inertia, entropy, inverse differential moment
 - Ref: <http://haralick.org/journals/TexturalFeatures.pdf>



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Invariances

- All features are function of the distance d and the orientation β
- Rotation?
 - Average on all directions
- Scaling?
 - Multi-resolutions



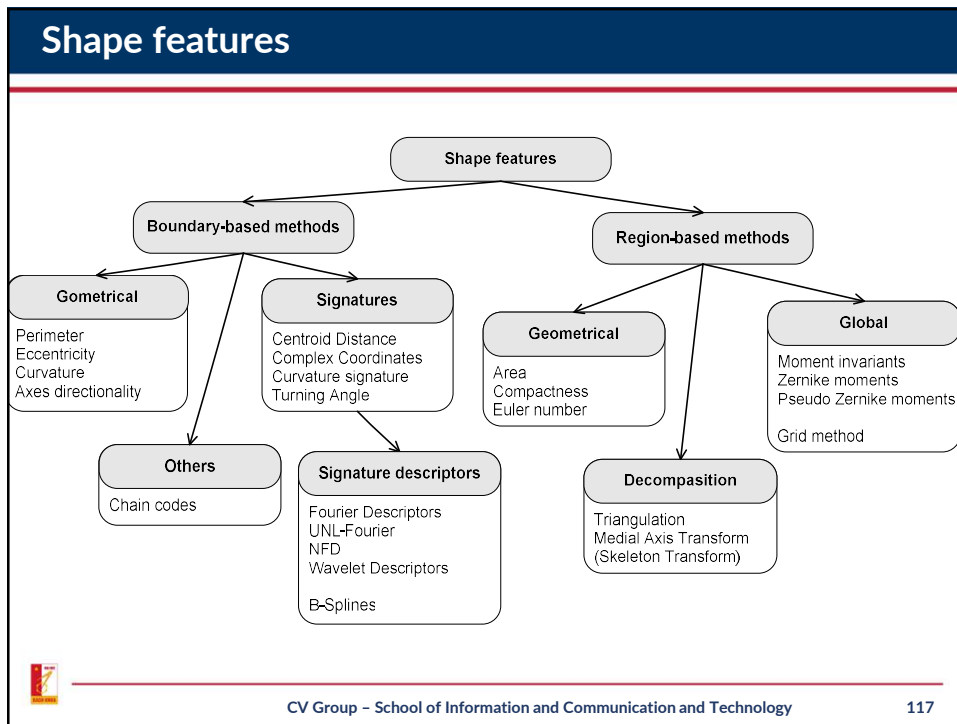
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Shape features

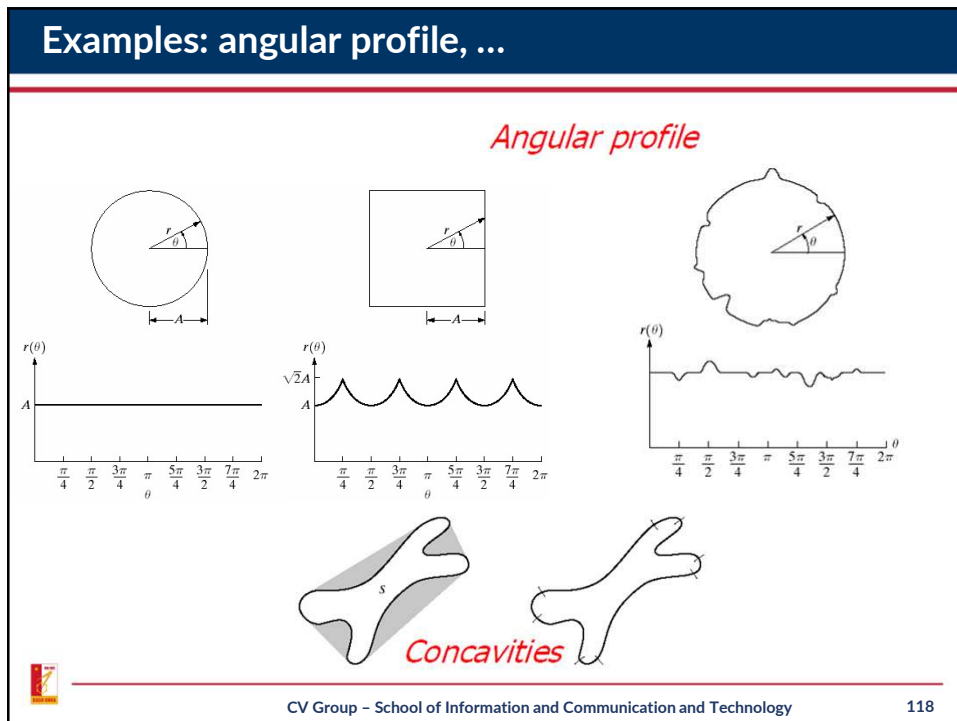
- Contour-based features:
 - Chain coding, polygon approximation, geometric parameters, angular profile, surface, perimeter, ...
- Region based:
 - Invariant moments, ...



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Examples : Image moments

- Moment

$$M_{ij} = \sum_x \sum_y x^i y^j I(x, y)$$

$M_{0,0}$ = area of the region D

- Central moments:

$(M_{0,1}, M_{1,0})$ = centroid of D

$$\mu_{pq} = \sum_x \sum_y (x - \bar{x})^p (y - \bar{y})^q I(x, y)$$

Invariant to translation

$$\bar{x} = \frac{M_{10}}{M_{00}} \quad \bar{y} = \frac{M_{01}}{M_{00}}$$



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Invariant moments (Hu's moments)

invariant to translation, scale, and rotation, and reflection

$$I_1 = \eta_{20} + \eta_{02}$$

$$I_2 = (\eta_{20} - \eta_{02})^2 + 4\eta_{11}^2$$

$$I_3 = (\eta_{30} - 3\eta_{12})^2 + (3\eta_{21} - \eta_{03})^2$$

$$I_4 = (\eta_{30} + \eta_{12})^2 + (\eta_{21} + \eta_{03})^2$$

$$I_5 = (\eta_{30} - 3\eta_{12})(\eta_{30} + \eta_{12})[(\eta_{30} + \eta_{12})^2 - 3(\eta_{21} + \eta_{03})^2] + (3\eta_{21} - \eta_{03})(\eta_{21} + \eta_{03})[3(\eta_{30} + \eta_{12})^2 - (\eta_{21} + \eta_{03})^2]$$

$$I_6 = (\eta_{20} - \eta_{02})[(\eta_{30} + \eta_{12})^2 - (\eta_{21} + \eta_{03})^2]$$

Change for image reflection

$$I_7 = (3\eta_{21} - \eta_{03})(\eta_{30} + \eta_{12})[(\eta_{30} + \eta_{12})^2 - 3(\eta_{21} + \eta_{03})^2] + (\eta_{30} - 3\eta_{12})(\eta_{21} + \eta_{03})[3(\eta_{30} + \eta_{12})^2 - (\eta_{21} + \eta_{03})^2]$$

$$\eta_{ij} = \frac{\mu_{ij}}{\mu_{00}^{(1+\frac{i+j}{2})}}$$



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Examples : Hu's moments

6 images and their Hu Moments

id	Image	H[0]	H[1]	H[2]	H[3]	H[4]	H[5]	H[6]
K0	K	2.78871	6.50638	9.44249	9.84018	-19.593	-13.1205	19.6797
S0	S	2.67431	5.77446	9.90311	11.0016	-21.4722	-14.1102	22.0012
S1	S	2.67431	5.77446	9.90311	11.0016	-21.4722	-14.1102	22.0012
S2	S	2.65884	5.7358	9.66822	10.7427	-20.9914	-13.8694	21.3202
S3	S	2.66083	5.745	9.80616	10.8859	-21.2468	-13.9653	21.8214
S4	S	2.66083	5.745	9.80616	10.8859	-21.2468	-13.9653	-21.8214



<https://www.learnopencv.com/wp-content/uploads/2018/12/HuMoments-Shape-Matching.png>

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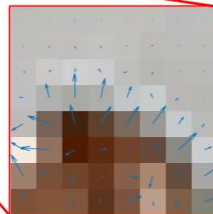
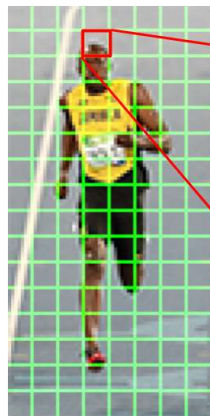
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HOG (Histogram of Oriented Gradients)

Idea: appearance of objects in an image can be described using the **distribution of edge directions**

- **HOG:** [Dalal, N., and Triggs, B. \(2005\). Histograms of oriented gradients for human detection, CVPR2005](#)



2	3	4	4	3	4	2	2
5	11	17	13	7	9	3	4
11	21	23	27	22	17	4	6
23	99	165	135	85	32	26	2
91	155	133	136	144	152	57	28
98	196	76	38	26	60	170	51
165	60	60	27	77	85	43	136
71	13	34	23	108	27	48	110

Gradient Magnitude

80	36	5	10	0	64	90	73
37	9	9	179	78	27	169	166
87	136	173	39	102	163	152	176
76	13	1	168	159	22	125	143
120	70	14	150	145	144	145	143
58	86	119	98	100	101	133	113
30	65	157	75	78	165	145	124
11	170	91	4	110	17	133	110

Gradient Direction

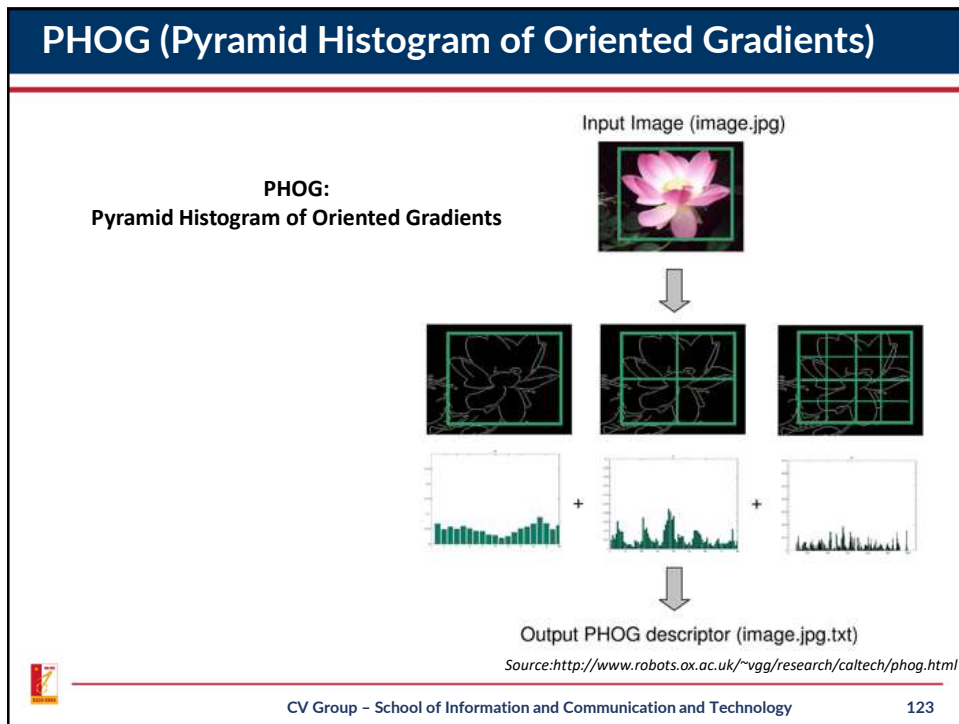
Source: LearnOpenCV.com



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Feature extraction

- Global features
- **Local features**
 - Interest point detector
 - Local descriptor

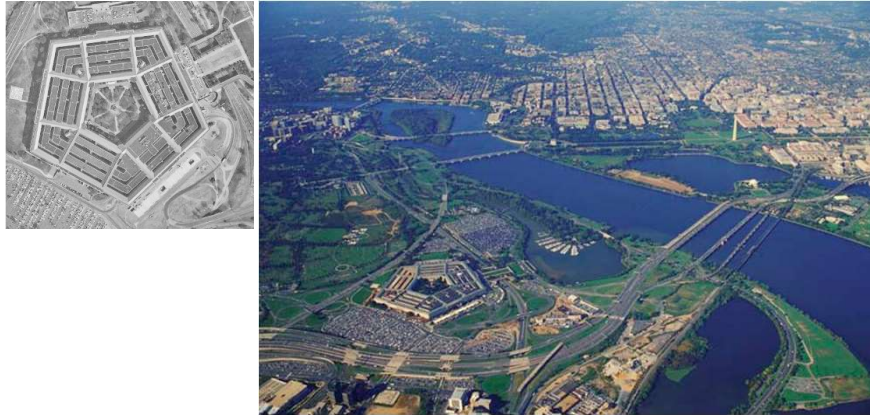
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Why local features?

- Image matching: a challenging problem



Source: CS131 - Juan Carlos Niebles and Ranjay Krishna

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Image matching



by [Diva Sian](#)

the Trevi Fountain in Rome



by [swashford](#)



by [scgbt](#)

Slide credit: Steve Seitz

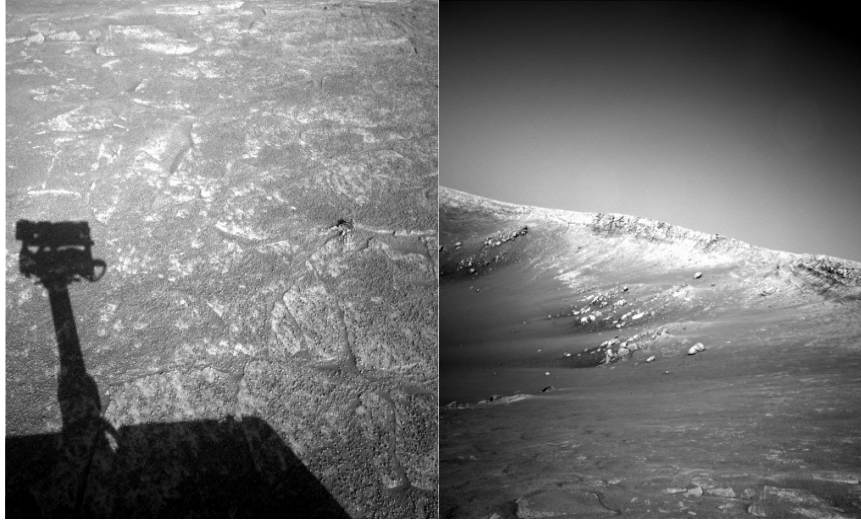


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Harder Still?



NASA Mars Rover images

Slide credit: Steve Seitz

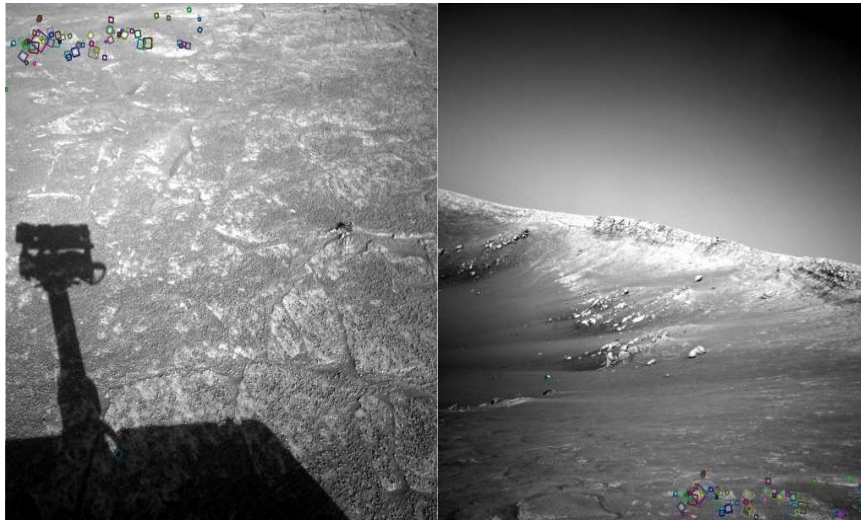


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Answer Below (Look for tiny colored squares)



NASA Mars Rover images with SIFT feature matches (Figure by Noah Snaveley)

Slide credit: Steve Seitz



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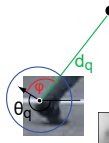
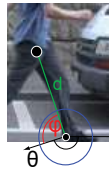
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Motivation for using local features

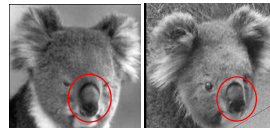
- Global representations have major limitations
- Instead, describe and match only local regions
- Increased robustness to
 - Occlusions



- Articulation



- Intra-category variations



Source: CS131 - Juan Carlos Niebles and Ranjay Krishna

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Local features and applications

- Partial search
- Object recognition/ detection



Sivic and Zisserman, 2003



D. Lowe 2002



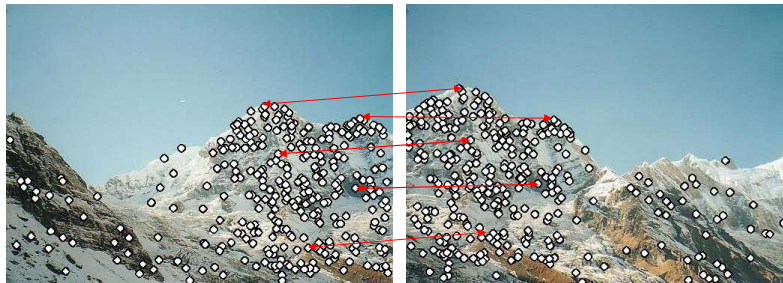
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Local features and applications

- Panorama:
 - Detect feature points in both images
 - Find corresponding pairs



[Darya Frolova and Denis Simakov]

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Local features and applications

- Panorama:
 - Detect feature points in both images
 - Find corresponding pairs
 - Use these pairs to align images



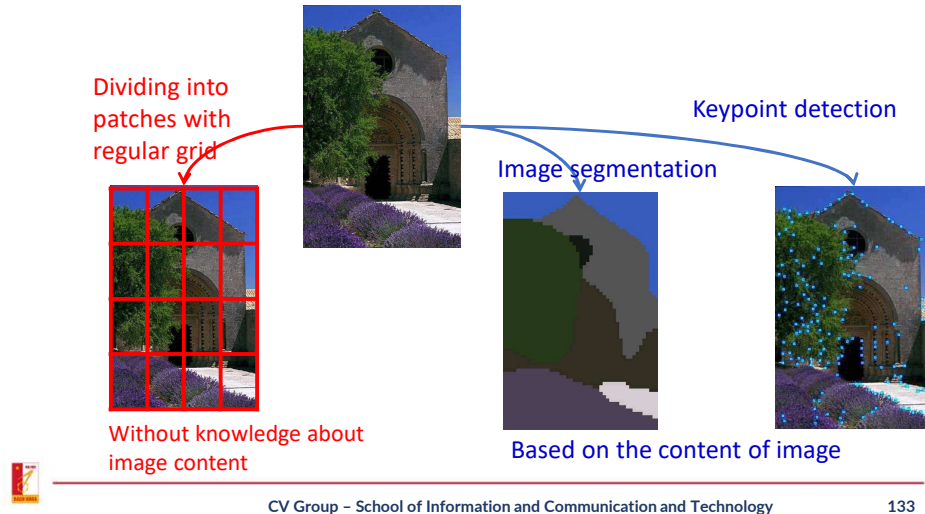
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Local feature extraction

- Local features: how to determine image patches / local regions



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Common Requirements

- Problem 1:
 - Detect the same point *independently* in both images



No chance to match!

- Problem 2:
 - For each point correctly recognize the corresponding one



We need a reliable and distinctive **descriptor**!

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Requirements- Summary

- Region extraction needs to be **repeatable** and **accurate**
 - **Invariant** to translation, rotation, scale changes
 - **Robust** or **covariant** to out-of-plane ($\approx$ affine) transformations
 - **Robust** to lighting variations, noise, blur, quantization
- **Locality**: Features are local, therefore robust to occlusion and clutter
- **Quantity**: We need a sufficient number of regions to cover the object
- **Distinctiveness**: The regions should contain “interesting” structure
- **Efficiency**: Close to real-time performance



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Main questions

- **Where** will the interest points come from?
 - What are salient features that we'll *detect* in multiple views?
- How **to describe** a local region?
- How to **establish correspondences**, i.e., compute matches?



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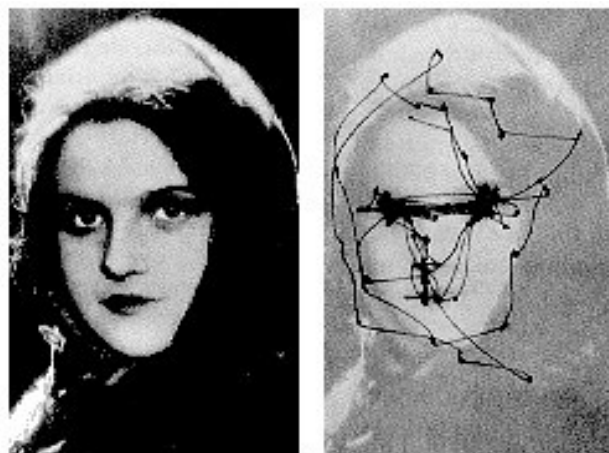
Feature extraction

- Global features
- **Local features**
 - Interest point detector
 - Local descriptor
 - Matching



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Interest points: why and where?



Yarbus eye tracking

Source : Derek Hoiem, Computer Vision, University of Illinois.



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Keypoint Localization

- Goals:

- Repeatable detection
- Precise localization
- Interesting content

⇒ *Look for two-dimensional signal changes*



Slide credit: Bastian Leibe

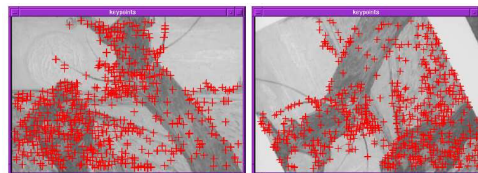


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Finding Corners



- Key property:

- In the region around a corner, image gradient has two or more dominant directions

- Corners are *repeatable* and *distinctive*

C.Harris and M.Stephens. "[A Combined Corner and Edge Detector.](#)" *Proceedings of the 4th Alvey Vision Conference*, 1988.

Slide credit: Svetlana Lazebnik



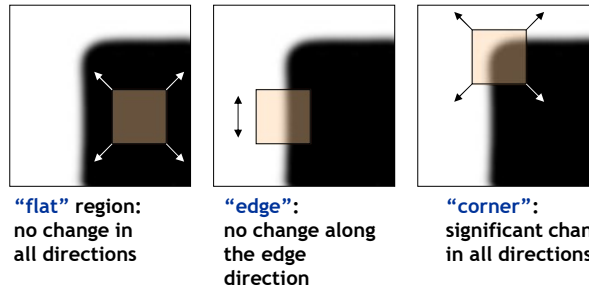
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Corners as distinctive interest points

- Design criteria
 - We should easily recognize the point by looking through a small window (*locality*)
 - Shifting the window in *any direction* should give a *large change* in intensity (*good localization*)



Slide credit: Alyosha Efros



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Corners versus edges

	$\sum I_x^2 \rightarrow \text{Large}$ $\sum I_y^2 \rightarrow \text{Large}$	Corner
	$\sum I_x^2 \rightarrow \text{Small}$ $\sum I_y^2 \rightarrow \text{Large}$	Edge
	$\sum I_x^2 \rightarrow \text{Small}$ $\sum I_y^2 \rightarrow \text{Small}$	Nothing



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Harris detector formulation

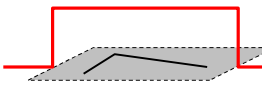
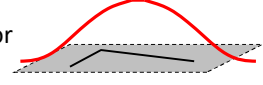
Change of intensity for the shift $[u, v]$:

$$E(u, v) = \sum_{x, y} w(x, y) [I(x + u, y + v) - I(x, y)]^2$$

Window
function

Shifted
intensity

Intensity

Window function $w(X, y) =$  or 
1 in window, 0 outside Gaussian



Source: R. Szeliski

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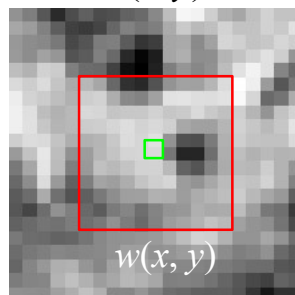
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Corner Detection by Auto-correlation

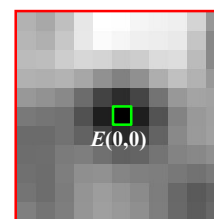
Change in appearance of window $w(x, y)$ for shift $[u, v]$:

$$E(u, v) = \sum_{x, y} w(x, y) [I(x + u, y + v) - I(x, y)]^2$$

$I(x, y)$



$E(u, v)$



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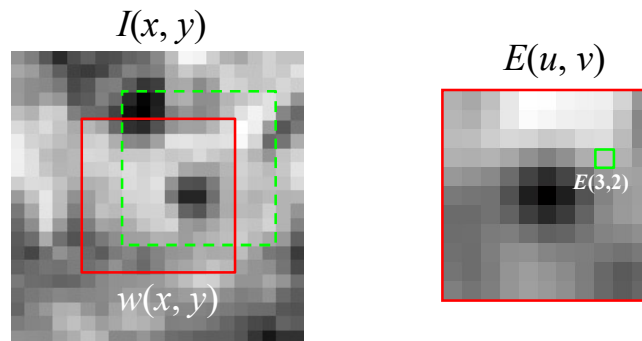
144

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Corner Detection by Auto-correlation

Change in appearance of window $w(x,y)$ for shift $[u,v]$:

$$E(u, v) = \sum_{x,y} w(x, y) [I(x+u, y+v) - I(x, y)]^2$$



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Harris detector formulation

- This measure of change can be approximated by:

$$E(u, v) \approx [u \ v] M \begin{bmatrix} u \\ v \end{bmatrix}$$

where M is a 2×2 matrix computed from image derivatives:

$$M = \sum_{x,y} w(x, y) \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix}$$

↑
Sum over image region – the area we are checking for corner

← Gradient with respect to x ,
times gradient with respect to y

$$M = \begin{bmatrix} \sum I_x I_x & \sum I_x I_y \\ \sum I_x I_y & \sum I_y I_y \end{bmatrix} = \sum \begin{bmatrix} I_x \\ I_y \end{bmatrix} [I_x \ I_y]$$



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What does this matrix reveal?

- First, let's consider an axis-aligned corner:

$$M = \begin{bmatrix} \sum I_x^2 & \sum I_x I_y \\ \sum I_x I_y & \sum I_y^2 \end{bmatrix} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$$



- This means:
 - Dominant gradient directions align with x or y axis
 - If either λ is close to 0, then this is not a corner, so look for locations where both are large.

Slide credit: David Jacobs



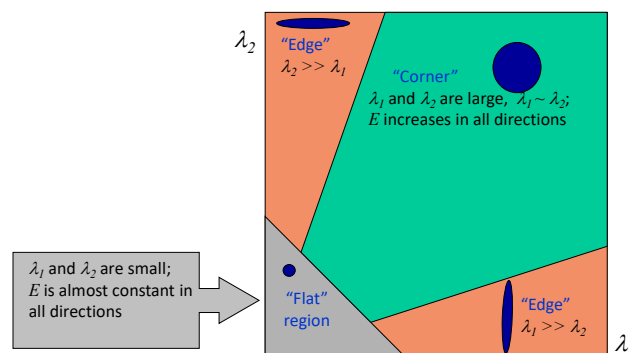
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Interpreting the eigenvalues

- Classification of image points using eigenvalues of M :



Slide credit: Kristen Grauman



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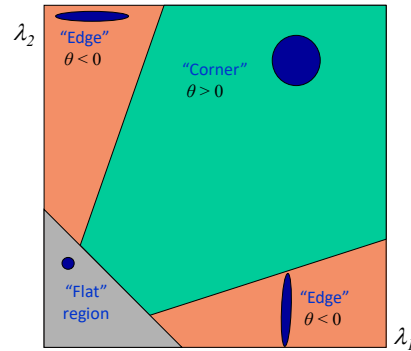
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Corner response function

$$\theta = \det(M) - \alpha \text{trace}(M)^2$$

- Fast approximation
 - Avoid computing the eigenvalues
 - α : constant (0.04 to 0.06)



Slide credit: Kristen Grauman

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Summary: Harris Detector [Harris88]

- Compute second moment matrix (autocorrelation matrix)

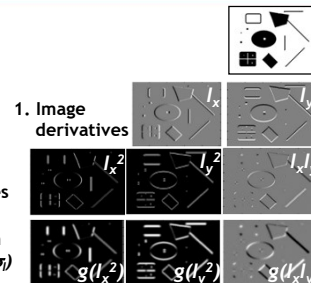
$$M(\sigma_I, \sigma_D) = g(\sigma_I) * \begin{bmatrix} I_x^2(\sigma_D) & I_x I_y(\sigma_D) \\ I_x I_y(\sigma_D) & I_y^2(\sigma_D) \end{bmatrix}$$

- Compute corner response

4. Cornerness function - two strong eigenvalues

$$\begin{aligned} \theta &= \det[M(\sigma_I, \sigma_D)] - \alpha [\text{trace}(M(\sigma_I, \sigma_D))]^2 \\ &= g(I_x^2)g(I_y^2) - [g(I_x I_y)]^2 - \alpha [g(I_x^2) + g(I_y^2)]^2 \end{aligned}$$

5. Perform non-maximum suppression



C.Harris and M.Stephens. "A Combined Corner and Edge Detector." *Proceedings of the 4th Alvey Vision Conference*: pages 147–151, 1988.

Slide credit: Krystian Mikolajczyk

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Harris Detector: Workflow



Slide adapted from Darya Frolova, Denis Simakov

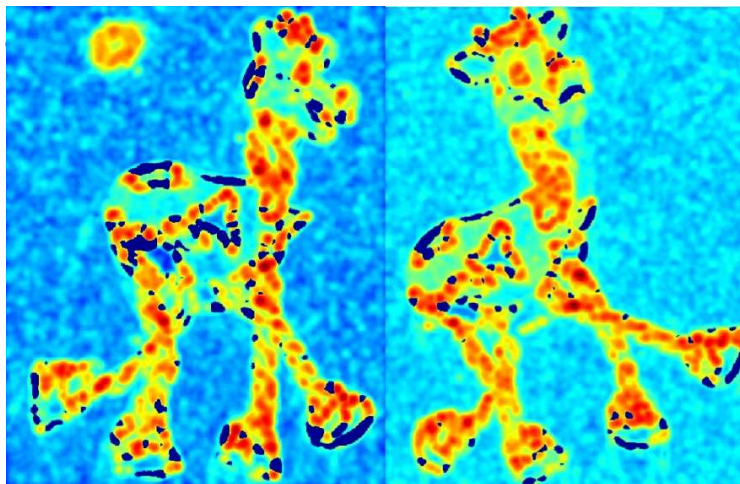


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Harris Detector: Workflow



Slide adapted from Darya Frolova, Denis Simakov



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Harris Detector: Workflow

Take points where $\theta > \text{threshold}$



Slide adapted from Darya Frolova, Denis Simakov



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Harris Detector: Workflow

Take only the local maxima of θ , where $\theta > \text{threshold}$



Slide adapted from Darya Frolova, Denis Simakov



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Harris Detector: Workflow

Resulting Harris points



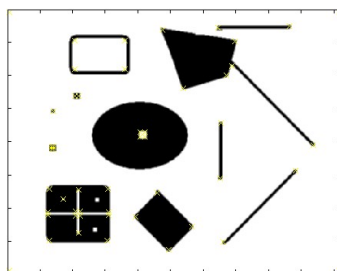
Slide adapted from Darya Frolova, Denis Simakov

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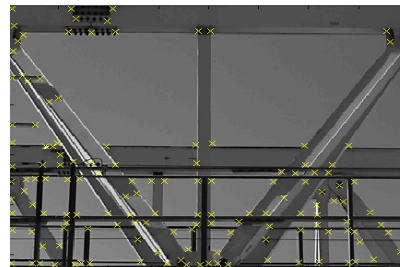
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Harris Detector – Responses [Harris88]



Effect: A very precise corner detector.



Slide credit: Krystian Mikołajczyk



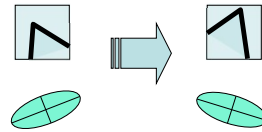
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Harris Detector: Properties

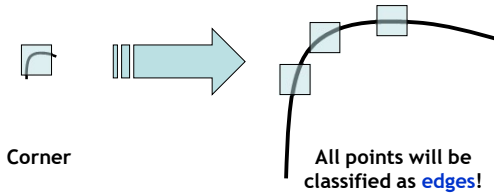
- Translation invariance
- Rotation invariance?



Ellipse rotates but its shape (i.e. eigenvalues) remains the same

Corner response θ is invariant to image rotation

- Scale invariance?



Corner

All points will be classified as *edges*!

Not invariant to image scale!



Slide credit: Kristen Grauman

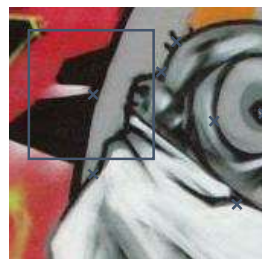
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Invariance

- Extract patch from each image individually



Slide adapted from T. Tuytelaars ECCV 2006 tutorial



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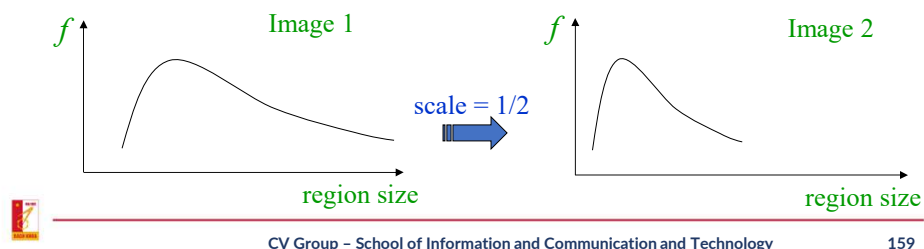
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Automatic scale selection

- Solution:
 - Design a function on the region, which is “scale invariant” (the same for corresponding regions, even if they are at different scales)

Example: average intensity. For corresponding regions (even of different sizes) it will be the same.

- For a point in one image, we can consider it as a function of region size (patch width)



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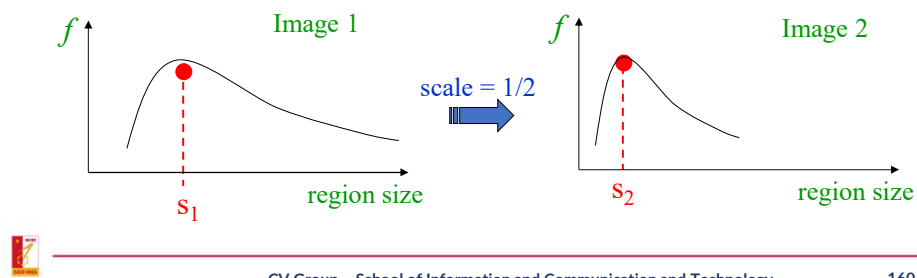
Automatic scale selection

- Common approach:

Take a local maximum of this function

Observation: region size, for which the maximum is achieved, should be *invariant* to image scale.

Important: this scale invariant region size is found in each image **independently!**



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Automatic Scale Selection



$$f(I_{i_1 \dots i_m}(x, \sigma)) = f(I_{i_1 \dots i_m}(x', \sigma'))$$

Same operator responses if the patch contains the same image up to scale factor.



K. Grauman, B. Leibe

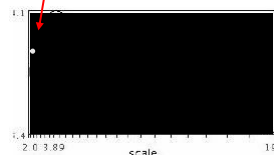
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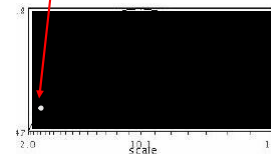
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Example

Function responses for increasing scale (scale signature)



$$f(I_{i_1 \dots i_m}(x, \sigma))$$



$$f(I_{i_1 \dots i_m}(x', \sigma))$$



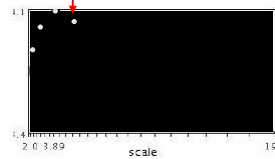
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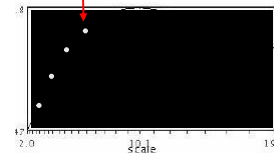
162

Example

Function responses for increasing scale (scale signature)



$$f(I_{i_1 \dots i_m}(x, \sigma))$$



$$f(I_{i_1 \dots i_m}(x', \sigma))$$



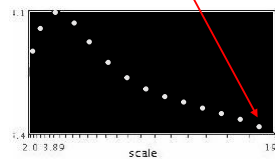
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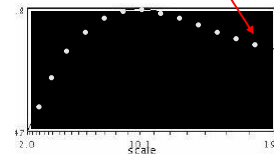
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Example

Function responses for increasing scale (scale signature)



$$f(I_{i_1 \dots i_m}(x, \sigma))$$



$$f(I_{i_1 \dots i_m}(x', \sigma))$$



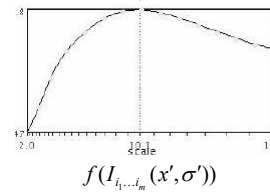
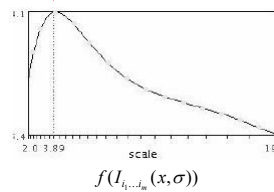
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Example

Function responses for increasing scale (scale signature)



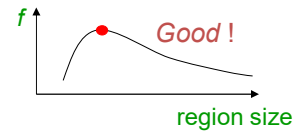
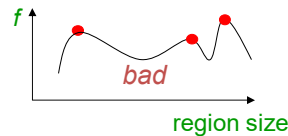
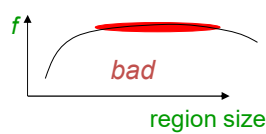
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Scale Invariant Detection

- A “good” function for scale detection:
has one stable sharp peak



- For usual images: a good function would be one which responds to contrast (sharp local intensity change)



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What is a useful signature function?

- Functions for determining scale $f = \text{Kernel} * \text{Image}$

Kernels:

$$L = \sigma^2 (G_{xx}(x, y, \sigma) + G_{yy}(x, y, \sigma))$$

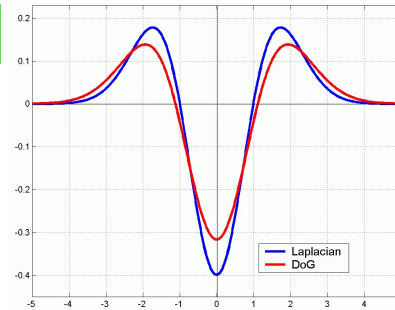
(Laplacian)

$$\text{DoG} = G(x, y, k\sigma) - G(x, y, \sigma)$$

(Difference of Gaussians)

where Gaussian

$$G(x, y, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x^2+y^2}{2\sigma^2}}$$



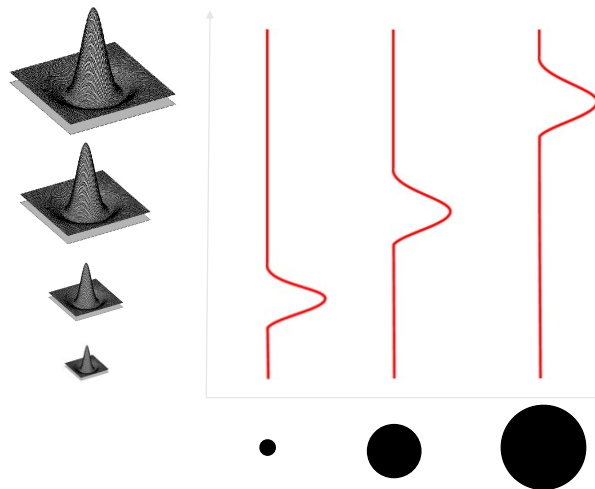
Note: both kernels are invariant to *scale* and *rotation*



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What is a useful signature function?

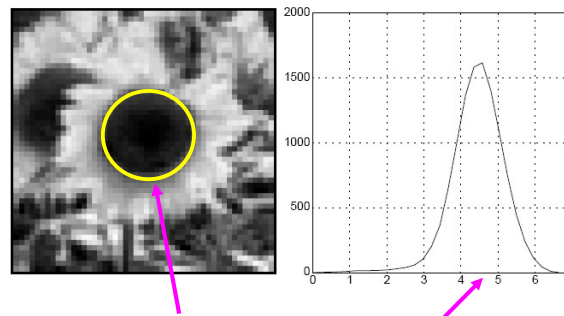
- Laplacian-of-Gaussian = “blob” detector



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Characteristic scale

- We define the **characteristic scale** as the scale that produces **peak of Laplacian response**



characteristic scale

T. Lindeberg (1998). "Feature detection with automatic scale selection." *IJCV* 30 (2): pp 77–116.

Source: Lana Lazebnik



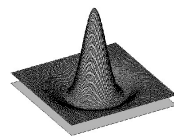
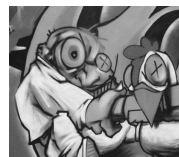
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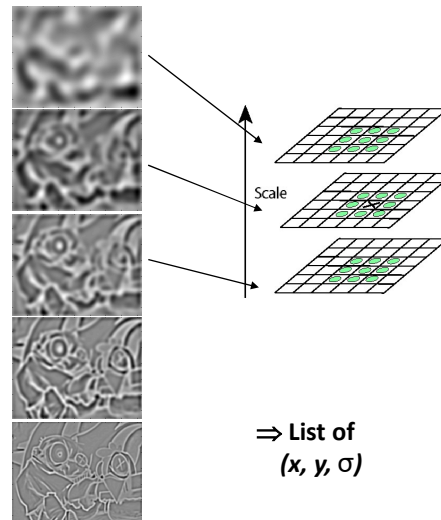
Laplacian-of-Gaussian (LoG)

- Interest points:
Local maxima in scale space of LoG



$$L_{xx}(\sigma) + L_{yy}(\sigma) = \sigma^3$$

σ^5
 σ^4
 σ^3
 σ^2
 σ



⇒ List of
(x, y, σ)



Source: K. Grauman, B. Leibe

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Alternative approach

- Approximate LoG with Difference-of-Gaussian (DoG):
 - 1. Blur image with σ Gaussian kernel
 - 2. Blur image with $k\sigma$ Gaussian kernel
 - 3. Subtract 2. from 1.

Small k gives a closer approximation to LoG, but usually we want to build a scale space quickly out of this. $k = 1.6$ gives an appropriate scale space, $k = \sqrt{2}$



Source: K. Grauman, B. Leibe

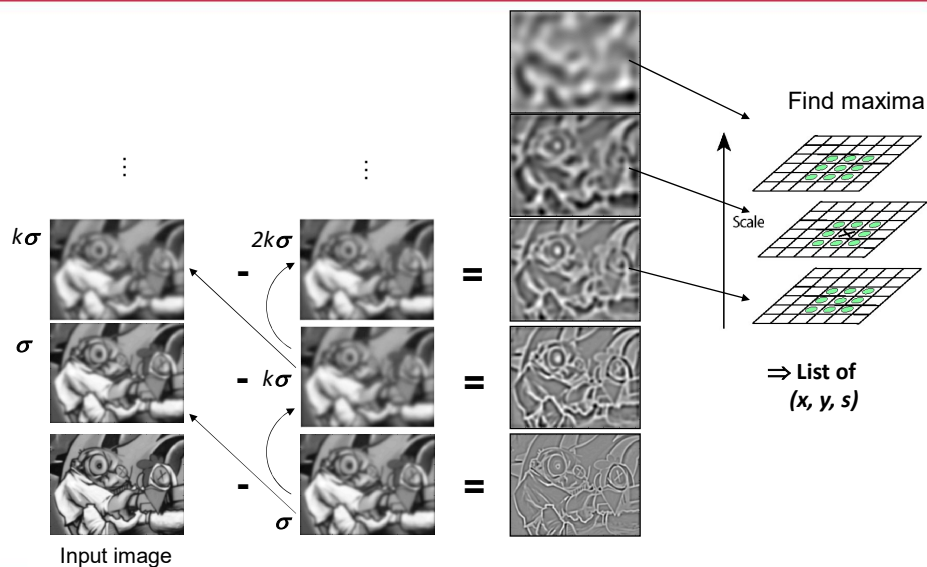


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Find local maxima in position-scale space of DoG



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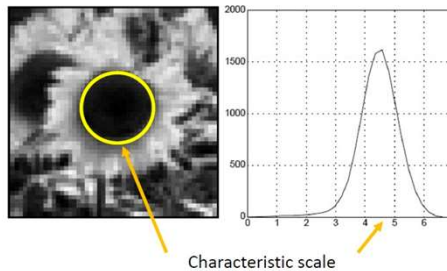
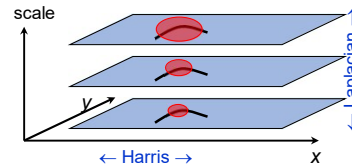
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Harris-Laplacian

• Harris-Laplacian¹

Find local maximum of:

- Harris corner detector in space (image coordinates)
- Laplacian in scale



¹ K.Mikolajczyk, C.Schmid. "Indexing Based on Scale Invariant Interest Points". ICCV 2001
² D.Lowe. "Distinctive Image Features from Scale-Invariant Keypoints". IJCV 2004

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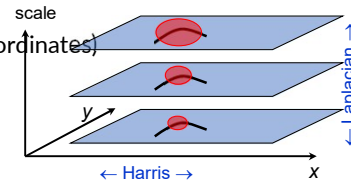
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Scale Invariant Detectors

• Harris-Laplacian¹

Find local maximum of:

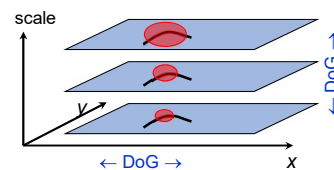
- Harris corner detector in space (image coordinates)
- Laplacian in scale



■ SIFT (D.Lowe)²

Find local maximum of:

- Difference of Gaussians in space and scale



¹ K.Mikolajczyk, C.Schmid. "Indexing Based on Scale Invariant Interest Points". ICCV 2001
² D.Lowe. "Distinctive Image Features from Scale-Invariant Keypoints". IJCV 2004

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DoG (SIFT) keypoint Detector

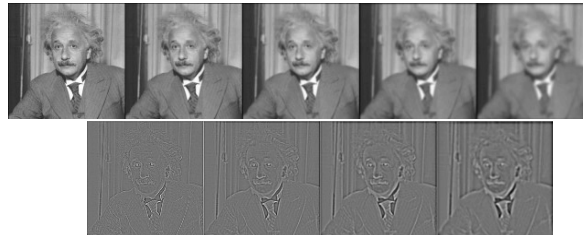
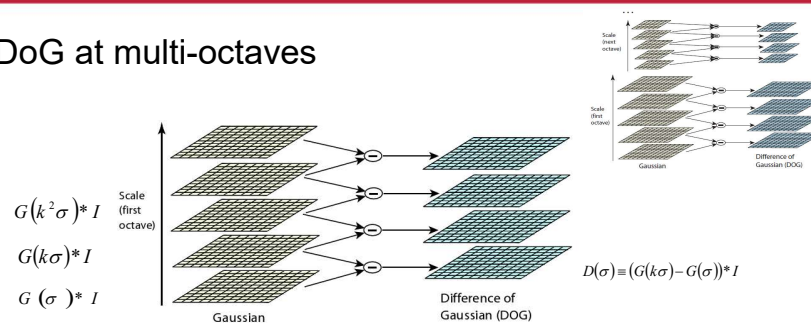
- DoG at multi-octaves
- Extrema detection in scale space
- Keypoint location
 - Interpolation
 - Removing instable points
- Orientation Assignment



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DoG (SIFT) Detector

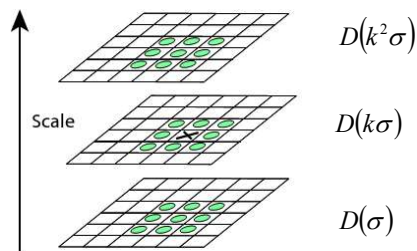
- DoG at multi-octaves



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DoG (SIFT) Detector

- Scale-Space Extrema Choose all extrema within 3x3x3 neighborhood



X is selected if it is larger or smaller than all 26 neighbors (its 8 neighbors in the current image and 9 neighbors each in the scales above and below)

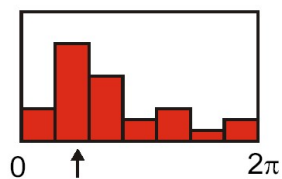
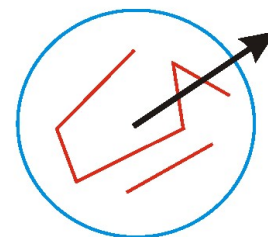


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DoG (SIFT) Detector

- Orientation assignment
 - Create histogram of local gradient directions at selected scale
 - Assign orientation at peak of smoothed histogram
- Each key specifies stable 2D coordinates

(x, y, scale, orientation)

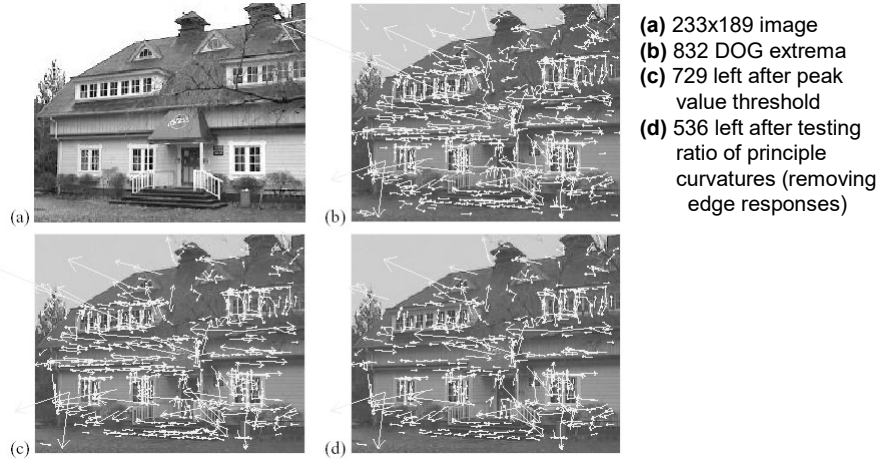


If 2 major orientations, use both.



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Example of keypoint detection



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DoG (SIFT) Detector



A SIFT keypoint : $\{x, y, \text{scale}, \text{dominant orientation}\}$

Source: [Distinctive Image Features from Scale-Invariant Keypoints](#) – IJCV 2004



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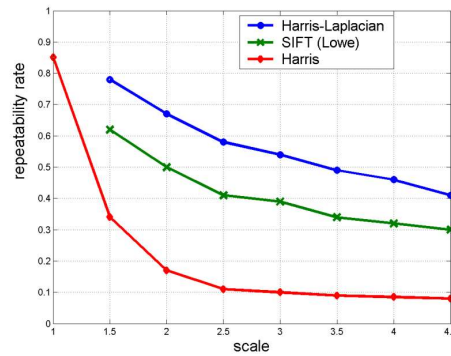
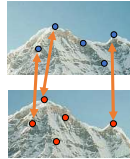
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Scale Invariant Detectors

- Experimental evaluation of detectors w.r.t. scale change

Repeatability rate:

$$\frac{\# \text{ correspondences}}{\# \text{ possible correspondences}}$$



K.Mikolajczyk, C.Schmid. "Indexing Based on Scale Invariant Interest Points". ICCV 2001



Slide credit: CS131 -Juan Carlos Niebles and Ranjay Krishna

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Many existing detectors available

- Hessian & Harris [Beaudet '78], [Harris '88]
 - Laplacian, DoG [Lindeberg '98], [Lowe '99]
 - Harris-/Hessian-Laplace [Mikolajczyk & Schmid '01]
 - Harris-/Hessian-Affine [Mikolajczyk & Schmid '04]
 - EBR and IBR [Tuytelaars & Van Gool '04]
 - MSER [Matas '02]
 - Salient Regions [Kadir & Brady '01]
 - Others...
- Those detectors have become a basic building block for many recent applications in Computer Vision.



Slide credit: Bastian Leibe

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Feature extraction

- Global features
- **Local features**
 - Interest point detector
 - Local descriptor
 - Matching



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Local Descriptor

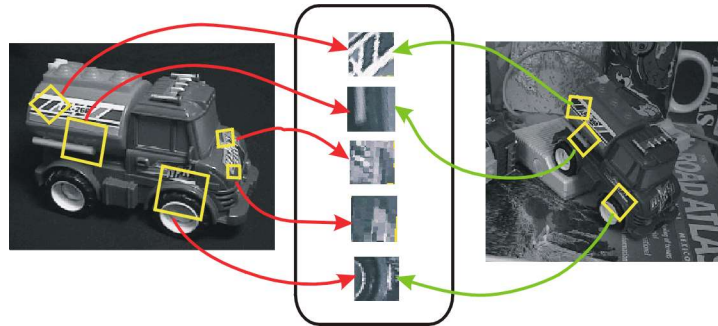
- Compact, good representation for local information
- Invariant
 - Geometric transformations: rotation, translation, scaling,...
 - Camera view change
 - Illumination
- Examples
 - SIFT, SURF([Speeded Up Robust Features](#)), [PCA-SIFT](#), ...
 - LBP, BRISK, MSER and FREAK, ...



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Invariant local features

- Image content is transformed into local feature coordinates that are invariant to translation, rotation, scale, and other imaging parameters



Following slides credit: CVPR 2003 Tutorial on **Recognition and Matching Based on Local Invariant Features** David Lowe



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Advantages of invariant local features

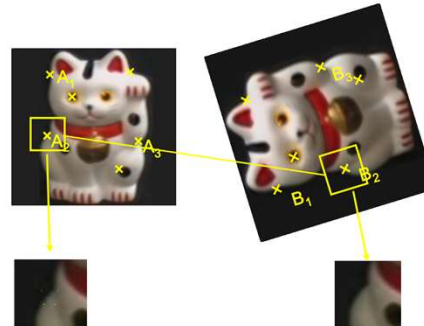
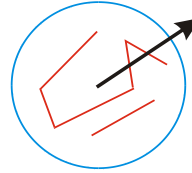
- **Locality:**
 - features are local, so robust to occlusion and clutter (no prior segmentation)
- **Distinctiveness:**
 - individual features can be matched to a large database of objects
- **Quantity:**
 - many features can be generated for even small objects
- **Efficiency:**
 - close to real-time performance
- **Extensibility:**
 - can easily be extended to wide range of differing feature types, with each adding robustness



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Becoming rotation invariant

- We are given a keypoint and its scale from DoG
- We will select a **characteristic orientation** for the keypoint (based on the most prominent gradient there)
- We will describe all features **relative** to this orientation



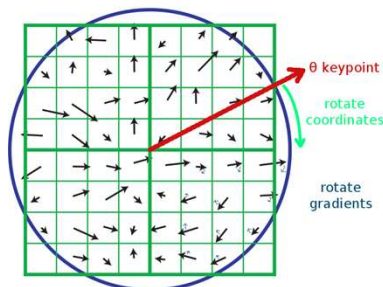
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SIFT descriptor formation

- Use the blurred image associated with the **keypoint's scale**
- Take image gradients over the keypoint neighborhood.
- In order to achieve orientation invariance, the **coordinates** of the descriptor and the **gradient orientations** are **rotated** relative to the keypoint orientation



Source: [Distinctive Image Features from Scale-Invariant Keypoints](http://campar.in.tum.de/twiki/pub/Chair/TeachingWs13TDCV/feature_descriptors.pdf) – IJCV 2004
http://campar.in.tum.de/twiki/pub/Chair/TeachingWs13TDCV/feature_descriptors.pdf



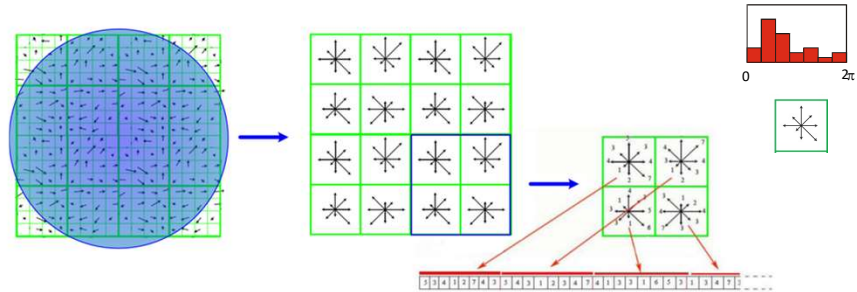
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SIFT descriptor formation

- Create array of orientation histograms (a 4x4 array is shown)
 - Each local orientation histogram has n orientation bins (here $n = 8$)
- The SIFT authors found that best results were with **8 orientation** bins per histogram and **a 4x4 histogram array** → a SIFT descriptor: vector of 128 values



SIFT: [Distinctive Image Features from Scale-Invariant Keypoints](#) – IJCV 2004

Image: Ashish A Gupta, PhD thesis 2013

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SIFT descriptor: more

- More about techniques used to SIFT become more stable and robust: see detail in section 6.2 at <https://www.cs.ubc.ca/~lowe/papers/ijcv04.pdf>



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SIFT

- Extraordinarily robust matching technique
 - Can handle changes in viewpoint: up to about 60 degree out of plane rotation
 - Can handle significant changes in illumination
 - Sometimes even day vs. night (below)
 - Fast and efficient—can run in real time



Steve Seitz

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Sensitivity to number of histogram orientations

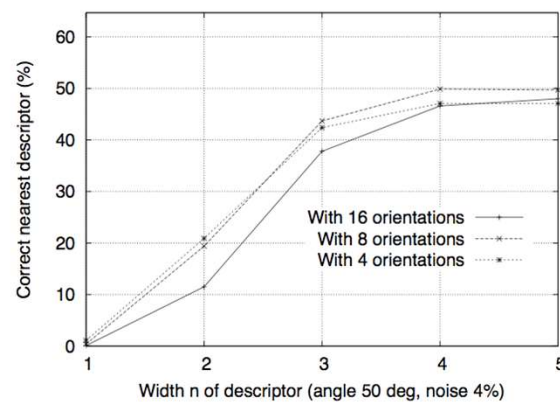


Figure 8: This graph shows the percent of keypoints giving the correct match to a database of 40,000 keypoints as a function of width of the $n \times n$ keypoint descriptor and the number of orientations in each histogram. The graph is computed for images with affine viewpoint change of 50 degrees and addition of 4% noise.



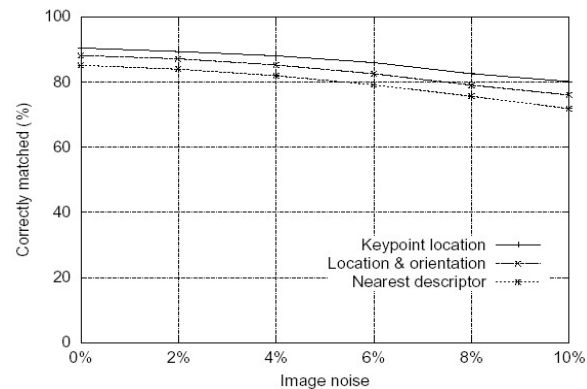
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Feature stability to noise

- Match features after random change in image scale & orientation, with differing levels of image noise
- Find nearest neighbor in database of 30,000 features



David G. Lowe, "Distinctive image features from scale-invariant keypoints," IJCV, 60, 2 (2004), pp. 91-110

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Feature stability to affine change

- Match features after random change in image scale & orientation, with 2% image noise, and affine distortion
- Find nearest neighbor in database of 30,000 features

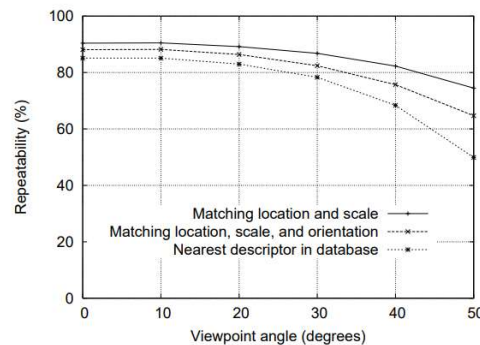


Figure 9: This graph shows the stability of detection for keypoint location, orientation, and final matching to a database as a function of affine distortion. The degree of affine distortion is expressed in terms of the equivalent viewpoint rotation in depth for a planar surface.

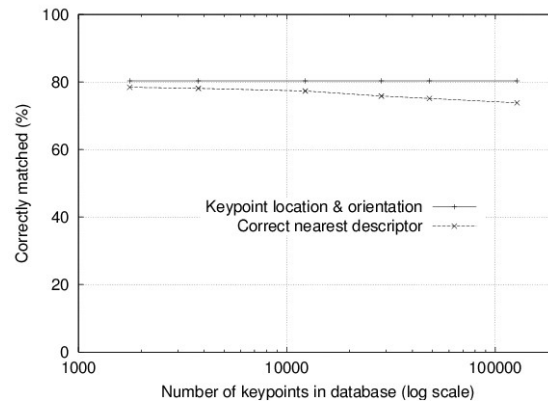
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Distinctiveness of features

- Vary size of database of features, with 30 degree affine change, 2% image noise
- Measure % correct for single nearest neighbor match

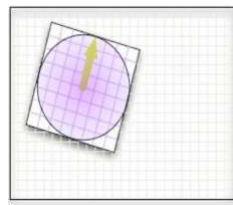


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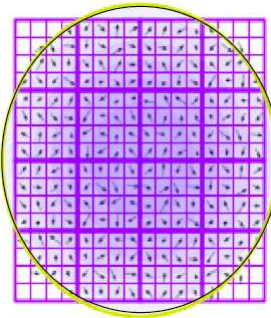
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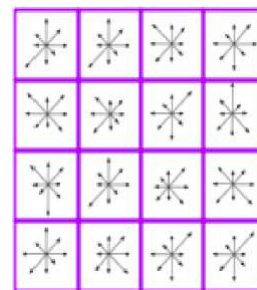
SIFT Keypoint Descriptor: summary



Blur the image using the scale of the keypoint (scale invariance)



Compute gradients in respect to the keypoint orientation (rotation invariance)



Compute orientation histogram in 8 directions over 4x4 sample regions

Source: [Distinctive Image Features from Scale-Invariant Keypoints](http://campar.in.tum.de/twiki/pub/Chair/TeachingWs13TDCV/feature_descriptors.pdf) – IJCV 2004
http://campar.in.tum.de/twiki/pub/Chair/TeachingWs13TDCV/feature_descriptors.pdf



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Other detectors and descriptors

Popular features: SURF, HOG, SIFT

http://campar.in.tum.de/twiki/pub/Chair/TeachingWs13TDCV/feature_descriptors.pdf

Summary some local features:

http://www.cse.iitm.ac.in/~vplab/courses/CV_DIP/PDF/Feature_Detectors_and_Descriptors.pdf



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Feature extraction

- Global features
- **Local features**
 - Interest point detector
 - Local descriptor
 - Matching



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Feature matching

Given a feature in I_1 , how to find the best match in I_2 ?

1. Define distance function that compares two descriptors
 - Use L1, L2, cosine, Mahalanobis,... distance
2. Test all the features in I_2 , find the one with min distance

OpenCV:

- Brute force matching
- Flann Matching: Fast Library for Approximate Nearest Neighbors [Muja and Lowe, 2009]

Marius Muja and David G Lowe. Fast approximate nearest neighbors with automatic algorithm configuration. In VISAPP (1), pages 331–340, 2009



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Feature matching

- How to define the difference between two features f_1, f_2 ?
 - Simple approach: use only distance value $d(f_1, f_2)$
 - → can give good score to very ambiguous matches
- Better approaches: add additional constraints
 - Radio of distance
 - Spatial constraints between neighborhood pixels
 - Fitting the transformation, then refine the matches (RANSAC)



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Feature matching

- Simple approach: use distance value $d(f_1, f_2)$
 - can give good score to very ambiguous matches



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Feature matching

- Better approaches: $\text{radio of distance} = d(f_1, f_2) / d(f_1, f_2')$
 - f_2 is best match to f_1 in I_2 ;
 - f_2' is 2nd best SSD match to f_1 in I_2
 - An ambiguous/bad match will have ratio close 1
 - Look for unique matches which have low ratio



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Ratio of distances reliable for matching

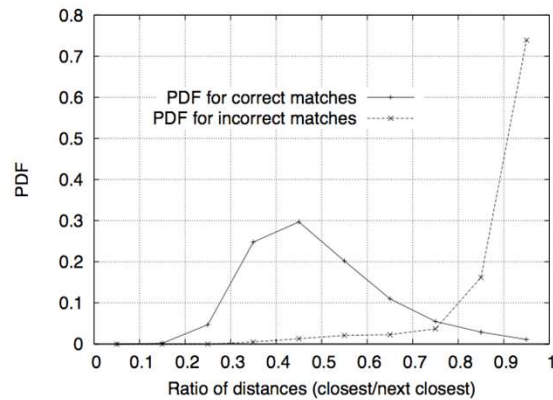


Figure 11: The probability that a match is correct can be determined by taking the ratio of distance from the closest neighbor to the distance of the second closest. Using a database of 40,000 keypoints, the solid line shows the PDF of this ratio for correct matches, while the dotted line is for matches that were incorrect.



David G. Lowe, "Distinctive image features from scale-invariant keypoints," IJCV, 60, 2 (2004), pp. 91-110

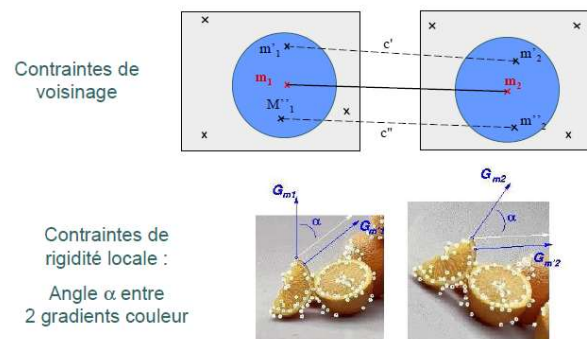
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Feature matching

- Better approaches: [Spatial constraints between neighborhood pixels](#)



Source: from slides of Valérie Gouet-Brunet

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Feature matching

- Better approaches: [fitting the transformation \(RANSAC alg.\)](#)

- Fitting 2D transformation matrix

- Six variables

- Each point give two equations
- → at least three points

- Least squares

$$H = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

- RANSAC: refinement of matches

- Compute error:

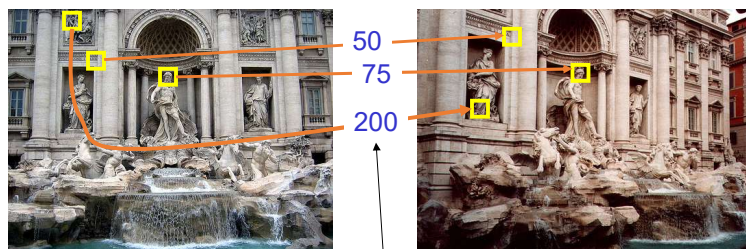
$$\left\| \begin{bmatrix} x_2 \\ y_2 \\ 1 \end{bmatrix} - H \begin{bmatrix} x_1 \\ y_1 \\ 1 \end{bmatrix} \right\|_2$$



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Evaluating the results

How can we measure the performance of a feature matcher?



Best feature distance

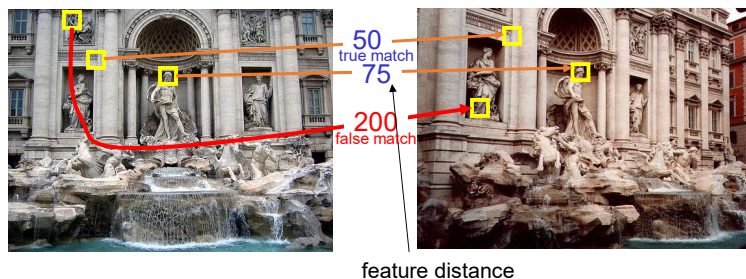


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True/false positives

The distance threshold affects performance

- **True positives** = # of detected matches that are correct
 - Suppose we want to maximize these—how to choose threshold?
- **False positives** = # of detected matches that are incorrect
 - Suppose we want to minimize these—how to choose threshold?



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Image matching

- How to define the distance between 2 images I_1, I_2 ?
 - Using global features: easy

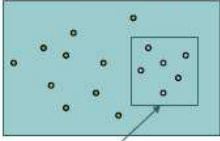
$$d(I_1, I_2) = d(\text{feature of } I_1, \text{feature of } I_2)$$
 - Using local features:
 - Voting strategy
 - Solving an optimization problem (time consuming)
 - Building a "global" feature from local features: BoW (bag-of-words, bag-of-features), VLAD, ..



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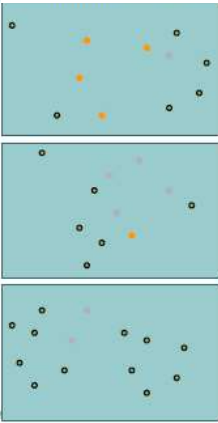
Voting strategy

Input image



Selected region
for query

Images in a database



- Distance faible $\leq \epsilon$
- Distance plus élevée $\leq \epsilon$
- Distance $> \epsilon$

The similarity between 2 images is based on **the number of matches**

Source: Modified from slides of Valérie Gouet-Brunet



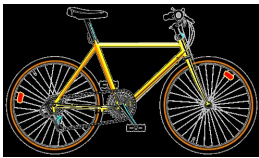
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Bag-of-words

- Local feature $\sim$ a word
- An image $\sim$ a document
- Apply a technique for textual document representation: vector model







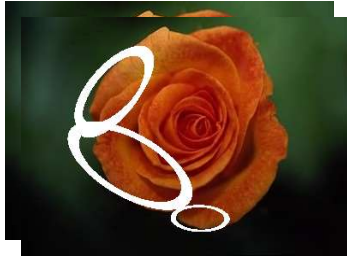


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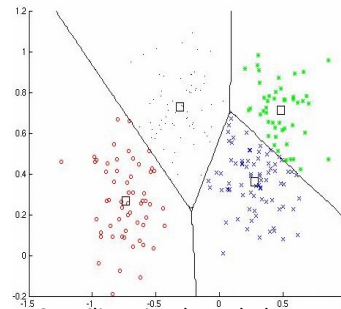
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Visual Vocabulary ...



1. Extracting local features from a set of images



2. Building visual vocabulary (dictionary) using a clustering method



3. An image is represented by a bag of words
→ can be represented by tf.idf vector



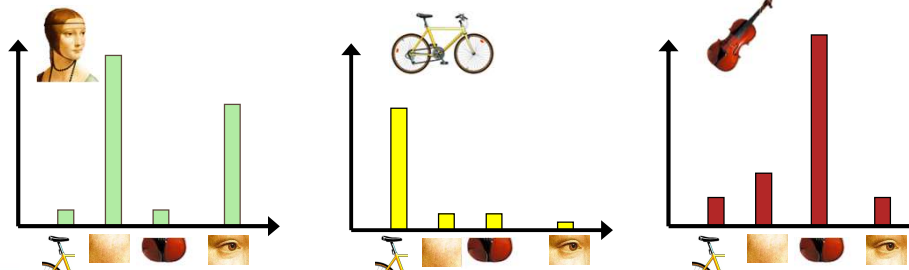
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Bag of words: outline

1. Extract features
2. Learn "visual vocabulary"
3. Quantize features using visual vocabulary
4. Represent images by frequencies of "visual words"



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Object detection/recognition/search



Sivic and Zisserman, 2003



Rothganger et al. 2003



Lowe 2002



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Object detection/recognition

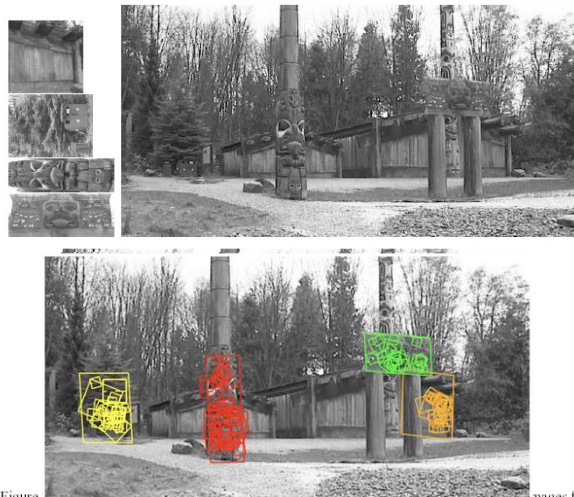
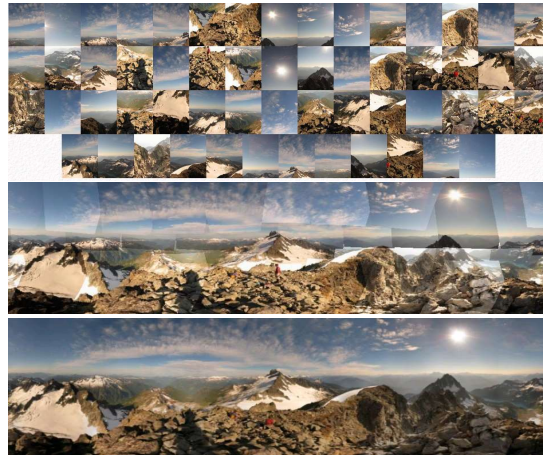


Figure 1. Images for locations are shown at the upper left and the 640x315 pixel test image taken from a different viewpoint is on the upper right. The recognized regions are shown on the lower image, with keypoints shown as squares and an outer parallelogram showing the boundaries of the training images under the affine transform used for recognition.



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Automatic mosaicing

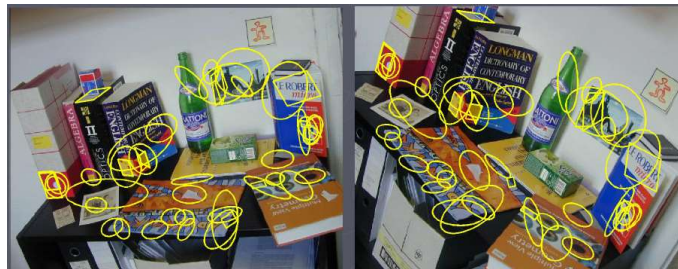


<http://www.cs.ubc.ca/~mbrown/autostitch/autostitch.html>



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Wide baseline stereo



[Image from T. Tuytelaars ECCV 2006 tutorial]



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References

- Lecture 5,6: CS231 - Juan Carlos Niebles and Ranjay Krishna, Stanford Vision and Learning Lab
- Vision par Ordinateur, Alain Boucher, IFI
- **SIFT**: [Keypoint detector \(ubc.ca\)](http://keypointdetector.ubc.ca/)
- **SURF**: <https://people.ee.ethz.ch/~surf/eccv06.pdf>
- **Harris corner detector**:
https://home.cis.rit.edu/~cnspci/references/dip/feature_extraction/harris1988.pdf
- **CBIR**: J. Sivic & A. Zisserman (2003). "[Video Google: A Text Retrieval Approach to Object Matching in Videos](#)" (PDF). *Proc. of ICCV*
- **HOG for Human detection**:
<https://lear.inrialpes.fr/people/triggs/pubs/Dalal-cvpr05.pdf>



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